

RAZOR: PRUNING REPLACEABLE EXPERTS IN LLMs

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[Code](#) [Models](#)

ABSTRACT

Mixture-of-experts (MoE) models activate only a few experts per token but store the entire expert pool. Pruning this pool requires identifying experts whose removal preserves model behavior. Routing frequency and output magnitude do not fully describe deletion damage, which also depends on how the surviving and replacement experts compensate for the removed output. We introduce RAZOR, a training-free pruning method based on consensus residuals, the deviations of expert outputs from their original weighted mixture. At a fixed layer input, these residuals give the exact output change for a single deletion under survivor renormalization and router refill. RAZOR aggregates this damage by conditional root mean square and selects experts under a layerwise budget using forward computation alone, without gradients, subset search, or recovery training. Against frequency, activation-norm, and REAP baselines on GLM-4.7-Flash and Qwen3.6-35B-A3B at 25% and 50% expert removal, it attains the highest macro average over nine reasoning-intensive tasks in all four model-budget settings, gaining 2.12–5.59 points over REAP and lowering reverse KL in all four. On DeepSeek-V4-Flash-0731 and Hy3, it also achieves the highest macro average among the three residual criteria. Local exactness does not guarantee better joint pruning. Generation analyses show changes in diversity, formatting, and termination despite higher task scores.

1 INTRODUCTION

Mixture-of-experts (MoE) models increase capacity while activating only a small subset of experts per token, yet sparse computation does not reduce the cost of storing the full expert pool (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Whole-expert pruning addresses this burden by reducing the pool itself (Lu et al., 2024; Muzio et al., 2024; Lasby et al., 2026; Liu et al., 2024). Our focus is pruning reasoning models while retaining performance on tasks that require multi-step problem solving. Changes to intermediate predictions can affect later steps, motivating an evaluation of both predictive fidelity and complete task outputs. At a fixed budget, we assess *functional replaceability*, the extent to which the surviving computation can compensate for a removed expert.

Common pruning scores characterize experts by selection frequency or output magnitude (Muzio et al., 2024; Jaiswal et al., 2025; Lasby et al., 2026). These statistics do not capture how expert outputs combine, even when the output norm is weighted by routing mass. A large contribution may be replaceable by the surviving mixture, whereas a small one may supply a component that the survivors cannot recover. Isolated importance therefore differs from deletion damage, which depends on the computation available after removal.

Deletion changes a routed mixture by renormalizing the survivors’ weights and, when the router refills the vacated slot, promoting an unselected expert. Figure 1 isolates both effects in a constructed single-token example. Without refill, panel (a) shows that removing the expert closest to the original mixture is less harmful than removing the one with the smallest output norm. With refill, panel (b) shows that a promoted expert can compensate for a removed balancing contribution and change the preferred deletion. Output magnitude, fixed-support damage, and refill-aware damage can therefore select different experts. More generally, identical gates and output norms can correspond to different least-harmful deletions (Proposition 1).

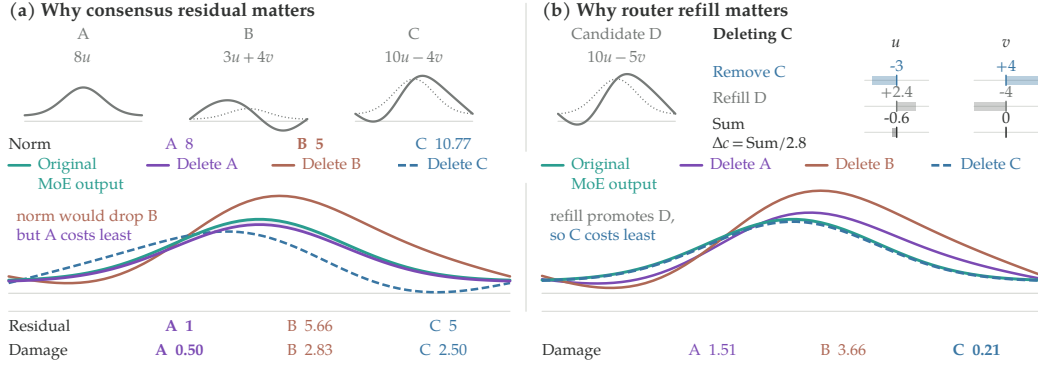


Figure 1: **Replaceability depends on output geometry and router refill.** A constructed single-token example without (a) and with (b) refill. Damage is L2 distance to the original mixture. Bold marks row minima. Appendix A gives the construction and plotting details.

We study three consensus-residual criteria that account for different parts of the fixed-input deletion counterfactual. The reference is the original weighted MoE output, which we call the *consensus* without assuming expert agreement. RCS measures each expert’s routing-weighted deviation from this mixture. RCS-LOO accounts for survivor renormalization, and RCS-Refill additionally includes router-selected replacement. We call the refill criterion with conditional root mean square (RMS) aggregation RAZOR. RCS-LOO and RCS-Refill recover exact single-deletion output changes under their respective fixed-input assumptions. Their comparison tests whether modeling these additional effects improves the pruned model. We retain the highest-scoring experts within each layer’s budget using forward computation, without gradients, subset search, or recovery training. Local change remains a surrogate for the model-level objective because joint deletions interact and altered activations propagate across layers.

Our benchmark suite spans six capability categories, namely competition mathematics, coding from function level to repository level, long-context inference, graduate-level knowledge, verifiable constraint following, and tool use. Scores assess final answers or executable outputs, providing evidence about task performance rather than the correctness of intermediate reasoning. We compare against pruning baselines on GLM-4.7-Flash and Qwen3.6-35B-A3B at 25% and 50% expert removal, where RAZOR has the highest nine-task macro average of all evaluated methods in every setting, exceeding REAP by 2.12–5.59 points, and also lowers reverse KL relative to REAP in all four. DeepSeek-V4-Flash-0731 and Hy3 extend the evaluation to two additional backbones, where RAZOR also achieves the highest nine-task macro average among the residual criteria.

The improvements have two limits. All four backbones fall below their unpruned macro scores at 50% removal, and better task performance does not consistently preserve generation behavior. On Qwen3.6-35B-A3B, diversity does not follow the task ranking, and RAZOR at 50% removal emits fewer stray closing delimiters than REAP while reaching the output cap more often than the unpruned model. These results motivate evaluating response behavior alongside accuracy when pruning reasoning models.

2 METHODOLOGY

2.1 PRUNING OBJECTIVE AND ROUTED MIXTURE

We retain B of E routed experts per layer, with $k \leq B < E$, without recovery training. Preserving the original output distribution at this budget is the goal. We use local Euclidean output change as a tractable surrogate, without assuming that minimizing it also minimizes model-level divergence.

For token representation x , the router selects k experts $S(x)$. Expert i produces $f_i(x)$ with normalized weight $w_i(x) \geq 0$, where $\sum_{i \in S(x)} w_i(x) = 1$. The routed output is

$$c(x) = \sum_{i \in S(x)} w_i(x) f_i(x). \quad (1)$$

We call this mixture the *consensus*, without assuming that its experts agree. Unchanged residual-connection, dense, and shared-expert branches cancel in the local comparison. A common routed-output scale does not change within-layer rankings (Appendix A.4), and Appendix B.3 lists each backbone’s scale.

The statistic $w_i \|f_i\|_2$ measures the magnitude of expert i ’s weighted output contribution. Deletion removes that contribution and renormalizes the survivors, so its local effect depends on information that even the complete set of gates and output norms cannot supply.

Proposition 1 (Magnitude does not identify the least harmful deletion). *Even with equal known gates, the expert-output norms do not in general determine which single-expert deletion minimizes the local output shift after survivor renormalization without refill.*

Proof. Consider equally weighted scalar outputs $(8, 3, 10)$ and $(8, 3, -10)$. Both have the same output norms and gates. Their original mixtures are 7 and $1/3$. The absolute shifts after deleting the first, second, or third expert are $(1/2, 2, 3/2)$ in the first case and $(23/6, 4/3, 31/6)$ in the second. The unique least harmful deletion is therefore the first expert in one case and the second in the other. A rule observing only the gates and norms cannot distinguish them. $\square$

Reweight output norms cannot recover the missing directional information. Given f_i and w_i , however, the shared vector c determines the survivor output.

2.2 FROM CONSENSUS RESIDUAL TO EXACT DELETION EFFECT

To isolate one deletion, we hold the layer input and expert outputs fixed and remove a selected expert with $w_i(x) < 1$. We first model router refill, then recover deletion without replacement as a special case.

Assumption 1 (Refill). Deleting $i \in S(x)$ promotes the highest-ranked unselected expert r under the router’s selection rule. Its nonnegative mixture score, divided by the original selected-score sum, defines the pseudo-weight $w_r(x)$. The surviving scores and this promoted score are then renormalized over their combined mass $1 - w_i(x) + w_r(x)$.

Assumption 2 (Fixed support). The surviving weights are renormalized over $S(x) \setminus \{i\}$ without promoting an unselected expert. Equivalently, $w_r(x) \equiv 0$ in Assumption 1.

We write $\tilde{c}^{-i}$ for the refilled mixture and c^{-i} for the fixed-support one. With a fixed ranking, the same highest-ranked unselected expert supplies the replacement for every single deletion at that token. The first comparison therefore uses the routed outputs and one additional expert output, whereas the second needs only the routed outputs. Neither requires evaluating a separate end-to-end pruned model for each expert.

Remark 1. Nonnegative $w_r(x)$ and $w_i(x) < 1$ ensure $D_i(x) \equiv 1 - w_i(x) + w_r(x) \geq 1 - w_i(x) > 0$. This argument does not require the selection ranking to match the ordering of mixture weights, which may differ when routing uses a selection-only correction bias (Appendix A.4).

Proposition 2 (Single-expert deletion under refill). *Under Assumption 1, the token-level deletion damage is*

$$\delta_i(x) \equiv \|c(x) - \tilde{c}^{-i}(x)\|_2 = \frac{\|w_i(x)r_i(x) - w_r(x)r_r(x)\|_2}{D_i(x)}, \quad (2)$$

where $r_j(x) = f_j(x) - c(x)$ is the consensus residual of expert j .

Proof. Suppressing x , the promoted slot carries mass w_r while the surviving selected mass is $1 - w_i$, so $\tilde{c}^{-i} = (c - w_i f_i + w_r f_r) / D_i$. Using $D_i - 1 = -w_i + w_r$,

$$c - \tilde{c}^{-i} = \frac{D_i c - c + w_i f_i - w_r f_r}{D_i} = \frac{w_i(f_i - c) - w_r(f_r - c)}{D_i}.$$

Taking norms yields the result. $\square$

Setting $w_r = 0$ recovers the fixed-support case in closed form.

Proposition 3 (Single-expert leave-one-out impact). *Under Assumption 2, the token-level deletion damage is*

$$\delta_i^{\text{loo}}(x) \equiv \|c(x) - c^{-i}(x)\|_2 = \frac{w_i(x)}{1 - w_i(x)} \|f_i(x) - c(x)\|_2. \quad (3)$$

Proof. Setting $w_r = 0$ in Proposition 2 gives denominator $1 - w_i$ and numerator $w_i \|f_i - c\|_2$. $\square$

These identities give replaceability a concrete form. Fixed-support damage is equivalently $w_i \|f_i - c^{-i}\|_2$, so distance to the survivor output measures how far the remaining computation is from reproducing the removed expert, and routing mass converts that distance into mixture change. Small damage is therefore ambiguous by construction, arising from either a small distance or a small routing mass. The LOO factor $1/(1 - w_i)$ corrects for self-inclusion in c and follows from the deletion counterfactual rather than serving as a tunable importance weight (Appendix A.2). Refill subtracts the promoted expert’s weighted residual from the removed expert’s residual and changes the normalizer. The promoted expert can therefore compensate for the removed contribution, though by an amount set by residual magnitudes and routing weights as well as direction. An exact match $w_i r_i = w_r r_r$ leaves the local mixture unchanged.

Exact single-deletion effects are not additive. Without refill, removing a routed subset $\mathcal{A} \subseteq S(x)$ with total weight $W_{\mathcal{A}} = \sum_{i \in \mathcal{A}} w_i < 1$ gives

$$c - c^{-\mathcal{A}} = \frac{\sum_{i \in \mathcal{A}} w_i (f_i - c)}{1 - W_{\mathcal{A}}}. \quad (4)$$

Residuals can reinforce or cancel, while removed mass changes the denominator (Appendix A.3). Scalar ranking does not optimize this joint effect, and refill does not restore additivity. Promoting several ranks at once couples the removed experts through both the numerator and the set-dependent normalizer. If $W_{\mathcal{A}} = 1$, no routing mass survives and the fixed-support counterfactual is undefined, even if $B \geq k$ experts remain in the layer. Deployment reselects top- k from the retained pool and propagates changed activations across layers, so matching its refill rule at a single token still does not guarantee checkpoint fidelity.

2.3 EXPERT SCORING AND PRUNING

A pruning rule requires one score per expert. RCS uses $w_i \|r_i\|_2$, RCS-LOO uses δ_i^{loo} , and RCS-Refill uses δ_i . Unless explicitly varied, all three aggregate their token quantities by root mean square (RMS) over calibration tokens routed to each expert, which weights variable damage above steady damage of the same average size. We write *RAZOR for RCS-Refill under conditional RMS*. This aggregation is an empirical choice rather than a consequence of the deletion identity. Appendices C.2 and C.3 analyze and test it, and Table 5 summarizes experimental coverage.

We retain the highest-scoring experts within each layer’s budget and deploy top- k routing over the retained pools. Appendix A.4 gives the implementation and pseudocode.

3 EXPERIMENTS

Four research questions narrow from what a pruned model predicts to why the scoring choices behind it matter.

- RQ1.** How well does pruning preserve output distributions across domains and routing conditions?
- RQ2.** How much downstream task performance does each criterion retain against prior scores, and does the ordering hold on further backbones?
- RQ3.** Which response properties shift even where task accuracy is retained?
- RQ4.** Do refinements to the local deletion score improve pruning, and how do scoring and aggregation choices affect expert selection and deletion-set interactions?

The experiments separately assess fidelity, task performance, response behavior, and the effects of individual scoring choices.

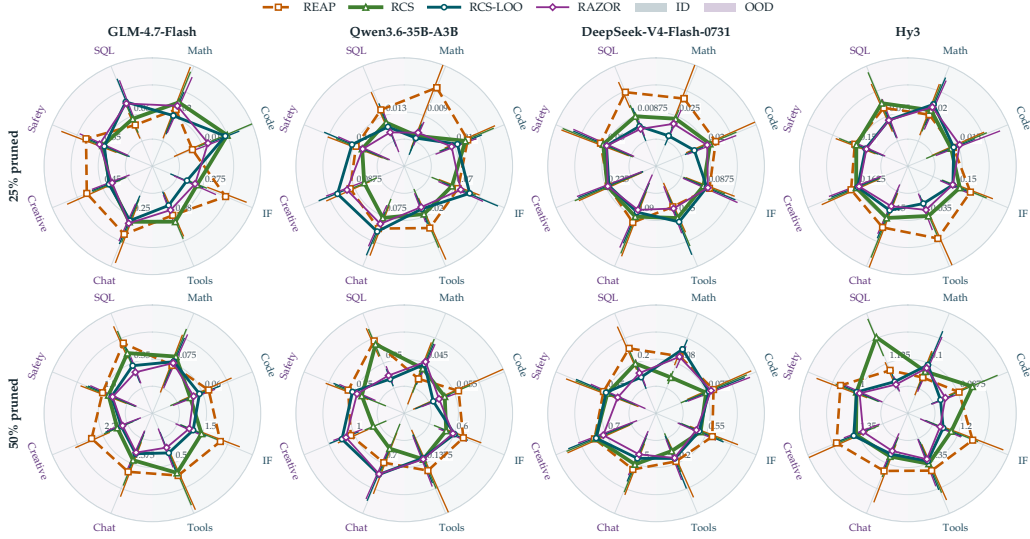


Figure 2: **Domain-wise reverse KL at 25% (top) and 50% (bottom) removal.** Values are in nats, with lower (inward) better. Compare methods along each spoke, not by polygon area. Whiskers show pointwise 95% paired-batch intervals. Shading distinguishes ID/OOD domains.

3.1 EXPERIMENTAL SETUP

Models and pruning methods. We compare RCS, RCS-LOO, and RAZOR on GLM-4.7-Flash, Qwen3.6-35B-A3B, DeepSeek-V4-Flash-0731, and Hy3 at 25% and 50% expert removal, without recovery training. All three use conditional RMS unless an aggregation ablation is specified, and each retains the highest-scoring experts in learned-router layers. Fixed-hash layers use the common pruning and route-remapping procedure in Appendix A.4. Matched checkpoint and component studies use GLM-4.7-Flash and Qwen3.6-35B-A3B, whose routers select 4/64 and 8/256 experts, respectively.

Baselines. We compare against routing frequency (Frequency), a conditional-mean adaptation of expert activation norm (EAN) (Jaiswal et al., 2025), and REAP (Lasby et al., 2026), which averages routing-weighted output norms conditional on use. Downstream benchmark comparisons against these three baselines use GLM-4.7-Flash and Qwen3.6-35B-A3B, while separate mask-based domain-wise and routing-stratified diagnostics compare against REAP on all four backbones (Appendices E.2 and E.3). REAP* denotes an external checkpoint calibrated with 24,576 samples, distinct from the matched REAP baseline.

Data and metrics. Calibration uses the 2,048-example, seven-domain RAZORCAL pool with model-specific chat templates. Predictive fidelity is reverse KL, $D_{KL}(q||p)$, from original (p) and pruned (q) predictions on shared reference prefixes. Assistant-token NLL gives $\Delta\text{NLL} = \text{NLL}_q - \text{NLL}_p$ and relative excess PPL, $\exp(\Delta\text{NLL}) - 1$. Fidelity is reported over eight axes, four covered by RAZORCAL (mathematics, code, instruction following, and tools) and four held out from it (chat, creative writing, safety, and SQL), so calibration-covered and held-out fidelity stay separable. This split is a property of the fidelity axes and does not carry over to the benchmark suite or the generation analyses. The matched component sweep pools tokens within axes and weights axes equally, whereas domain-wise and routing-stratified studies use grouped estimators.

The nine downstream tasks are AIME’26 (Mathematical Association of America, 2026), IFEval (Zhou et al., 2023), IFBench (Pyatkin et al., 2025), SuperGPQA (Du et al., 2025), BFCL v4 (Patil et al., 2025; Mao et al., 2025), LongBench v2 (Bai et al., 2025), HumanEval+ (Liu et al., 2023), LiveCodeBench 2026 (Jain et al., 2025), and SWE-bench Verified (Jimenez et al., 2024). They span competition mathematics, coding from function level to repository level, long-context inference, graduate-level knowledge, verifiable constraint following, and tool use. Each is scored on its final

Table 1: Nine-task benchmark results on GLM-4.7-Flash and Qwen3.6-35B-A3B at 25% and 50% removal. Scores are scaled by 100. Bold marks the best pruned score in each column within a model-budget block.

Model	Method	Math	Instruction Following				Know.	Tool	LC	Coding				Overall
		AIME'26	IFEval	IFBench	Avg	SuperGPQA	BFCL v4	LongB. v2	HE+	LCB'26	SWE	Avg	Avg	
GLM-4.7-Flash	0% —	31.7	79.7	40.0	59.9	38.5	65.3	25.3	79.9	37.4	5.2	40.8	44.8	
	Frequency	26.3	77.3	37.3	57.3	29.5	62.8	21.9	74.4	29.3	5.2	36.3	40.4	
	EAN	30.4	76.3	39.3	57.8	31.5	64.7	27.2	75.0	33.8	4.8	37.9	42.6	
	REAP*	30.0	74.9	36.7	55.8	32.6	64.9	27.2	75.0	30.6	3.6	36.4	41.7	
	25% REAP	30.8	74.9	37.3	56.1	33.7	65.0	27.0	75.0	34.3	3.6	37.6	42.4	
	RCS	29.6	77.1	36.0	56.6	33.3	65.7	26.0	73.2	33.8	4.2	37.1	42.1	
	RCS-LOO	31.3	76.7	39.7	58.2	34.3	64.8	27.4	75.6	34.8	4.2	38.2	43.2	
	RAZOR	33.3	78.2	40.3	59.3	34.2	65.6	28.5	76.2	39.8	4.6	40.2	44.5	
	Frequency	0.0	62.8	30.3	46.6	15.7	51.1	18.7	54.9	17.8	0.2	24.3	27.9	
	EAN	15.8	70.6	29.0	49.8	25.4	54.8	23.7	59.2	33.9	2.6	31.9	35.0	
	50% REAP	23.3	61.0	25.3	43.2	27.8	56.0	22.1	67.1	26.7	1.2	31.7	34.5	
	RCS	29.6	63.8	27.3	45.6	24.9	59.1	19.5	71.3	27.6	3.0	34.0	36.2	
	RCS-LOO	26.3	67.5	30.7	49.1	29.0	58.6	20.9	72.0	29.4	3.2	34.9	37.5	
	RAZOR	30.8	72.0	32.3	52.2	33.7	59.1	25.9	73.2	29.8	4.0	35.7	40.1	
Qwen3.6-35B-A3B	0% —	76.3	83.9	33.7	58.8	62.8	67.6	50.1	91.5	71.9	17.0	60.1	61.6	
	Frequency	75.0	79.1	34.3	56.7	51.4	67.2	45.9	90.9	70.1	12.0	57.7	58.4	
	EAN	75.4	81.7	32.3	57.0	57.4	65.8	48.7	92.7	64.8	16.6	58.0	59.5	
	25% REAP	77.1	79.5	33.7	56.6	57.0	65.8	48.8	90.9	70.9	14.6	58.8	59.8	
	RCS	76.7	80.0	38.0	59.0	57.3	66.3	48.5	92.1	71.9	18.0	60.7	61.0	
	RCS-LOO	76.3	80.6	34.7	57.7	57.1	66.4	49.7	92.7	72.6	17.4	60.9	60.8	
	RAZOR	77.5	81.9	38.3	60.1	57.8	67.0	49.9	93.3	72.6	20.2	62.0	62.1	
	Frequency	68.8	71.0	31.7	51.4	41.1	43.1	42.5	89.6	55.0	7.2	50.6	50.0	
	EAN	72.1	74.1	30.3	52.2	46.1	58.7	46.1	88.4	47.1	8.4	48.0	52.4	
	50% REAP	71.7	73.2	31.0	52.1	46.3	63.9	47.5	88.4	66.4	12.6	55.8	55.7	
	RCS	73.3	74.3	36.3	55.3	48.3	64.1	48.3	91.5	67.6	12.8	57.3	57.4	
	RCS-LOO	75.4	78.2	32.0	55.1	47.0	64.5	47.9	92.1	67.9	13.6	57.9	57.6	
	RAZOR	75.0	79.3	34.0	56.7	49.2	65.0	48.3	92.1	69.4	15.8	59.1	58.7	

answer or artifact, such as a numeric answer, a selected option, or code that passes tests, rather than on the intermediate steps. Appendix B.5 details task versions and score aggregation. Response analyses measure diversity, stray closing thinking delimiters, code-fence structure, length, and length-limit finishes. These are observable properties of the generated text, not measures of reasoning quality, and they qualify the task results without replacing them.

Hyperparameters and evaluation protocol. Matched comparisons use the same calibration inputs and layerwise budgets, with the external REAP* checkpoint reported separately. For downstream benchmarks, we use temperature 0, except for SWE-bench Verified, where we use 0.7. AIME'26 reports avg@8, and the other eight benchmarks report avg@3. Diversity and response-form diagnostics follow the separate protocol of Section 3.4. Context and output limits, the thinking setting, and cross-study matching are detailed in Appendices B.1–B.5. Diagnostic sampling and uncertainty are described in Appendices E.2–E.6.

3.2 PREDICTIVE FIDELITY (RQ1)

We first test whether scores based on local output change preserve the model’s predictive distribution after joint pruning. Figure 2 compares REAP and the three residual criteria per domain, Table 6 gives matched equal-axis GLM-4.7-Flash and Qwen3.6-35B-A3B estimates, and Figure 8 resolves the same comparison by routing concentration under both fidelity measures, separating calibration-covered from held-out domains. The three residual criteria use conditional RMS throughout, so contrasts among them isolate the token quantity, whereas REAP uses the conditional mean of its original definition, so comparisons against it vary aggregation as well. Lower reverse KL indicates less drift, and radar spokes are scaled independently from mostly nonzero origins, so methods are comparable on the same spoke and not by polygon area.

Obs 1. RAZOR reduces predictive drift, but KL and likelihood need not agree. Across the four matched GLM-4.7-Flash and Qwen3.6-35B-A3B settings, RAZOR has lower reverse KL than REAP in all four and than RCS-LOO in three (Table 6). A separate routing-stratified comparison across all four backbones illustrates the distinction between fidelity measures. At 50% removal, RAZOR lowers

Table 2: Nine-task benchmark results on DeepSeek-V4-Flash-0731 and Hy3 at 25% and 50% removal. These two backbones compare the residual criteria against the unpruned reference, without the Frequency, EAN, and REAP baselines of Table 1. Other conventions follow Table 1.

Model	Method	Math	Instruction Following			Know.	Tool	LC	Coding				Overall	
		AIME'26	IFEval	IFBench	Avg	SuperGPQA	BFCL v4	LongB. v2	HE+	LCB'26	SWE	Avg	Avg	
DeepSeek-V4-Flash-0731	0%	—	67.9	86.1	37.7	61.9	60.3	72.6	33.8	83.5	69.3	21.2	58.0	59.2
	25%	RCS	73.8	82.6	37.0	59.8	55.4	70.2	35.4	91.5	68.3	12.4	57.4	58.5
		RCS-LOO	76.7	83.7	38.7	61.2	55.6	70.4	35.2	89.6	68.8	18.8	59.1	59.7
		RAZOR	81.3	84.7	36.3	60.5	56.0	72.8	37.4	93.3	69.1	18.2	60.2	61.0
	50%	RCS	75.8	78.4	36.7	57.6	46.7	66.5	30.4	84.2	58.0	8.4	50.2	53.9
		RCS-LOO	76.7	79.8	39.0	59.4	46.5	68.1	29.6	85.4	58.4	10.4	51.4	54.9
		RAZOR	78.8	80.4	38.7	59.6	46.4	72.1	30.8	86.0	58.2	9.2	51.1	55.6
	Hy3	0%	—	82.9	90.4	46.7	68.6	64.7	64.2	50.0	90.8	68.3	15.4	58.2
25%		RCS	81.5	88.0	46.3	67.2	57.9	63.9	46.5	89.6	66.2	14.0	56.6	61.5
		RCS-LOO	83.0	89.1	46.0	67.6	58.3	64.2	47.3	90.2	65.7	13.4	56.4	61.9
		RAZOR	88.9	89.5	47.3	68.4	59.2	64.3	48.2	90.8	66.9	15.8	57.8	63.4
50%		RCS	86.8	81.6	42.7	62.2	45.5	59.3	42.7	87.2	58.6	10.0	51.9	57.2
		RCS-LOO	91.6	83.7	44.7	64.2	46.1	59.6	44.7	89.0	58.2	10.0	52.4	58.6
		RAZOR	92.3	84.3	45.0	64.7	46.3	60.2	44.3	90.2	61.7	11.2	54.4	59.5

KL relative to REAP in 73 of 80 model–domain–group–decile estimates, but keeps PPL closer to the original in only 47. The corresponding RCS-LOO counts are 72 and 47 (Appendix E.4). Reduced distributional drift therefore need not imply better preservation of reference-token likelihood.

Obs 2. KL improvements are not uniform across domains. Of eight domains, the counts where RAZOR lowers KL relative to REAP at 25% and 50% removal are, respectively, 5 and 7 for GLM-4.7-Flash, 6 and 5 for Qwen3.6-35B-A3B, 6 and 8 for DeepSeek-V4-Flash-0731, and 6 and 7 for Hy3. Yet Qwen3.6-35B-A3B’s equal-domain held-out mean at 50% is 0.2% higher than REAP’s, and the relative-change interval includes zero. Aggregate gains thus coexist with domain-level regressions. Appendix E.2 gives the corresponding RCS-LOO results. These domain-wise and routing-stratified views reweight overlapping predictions rather than provide independent replications.

3.3 DOWNSTREAM PERFORMANCE (RQ2)

Predictive fidelity on reference prefixes does not directly measure task performance under decoding. Tables 1 and 2 report the nine-task suite, with unpruned references at 0%. The first carries the baseline comparison, and the second tests whether the ordering among the residual criteria holds on two further routing architectures. Overall Avg weights all nine tasks equally, while the two Avg columns average instruction-following and coding tasks. Macro gaps quoted below use unrounded task-score averages.

Obs 3. RAZOR leads the baseline comparison in every setting where baselines were run. Against Frequency, EAN, and REAP on GLM-4.7-Flash and Qwen3.6-35B-A3B, RAZOR has the highest macro score in all four model–budget settings. Relative to REAP, it gains 2.12–5.59 points and wins all 36 paired task comparisons. RCS-LOO gains 0.80–3.01 points over REAP while losing three of those 36 comparisons. DeepSeek-V4-Flash-0731 and Hy3 extend the comparison among residual criteria, with the unpruned models as references and no Frequency, EAN, or REAP benchmark runs. RAZOR leads RCS-LOO by 0.74–2.58 points across all eight settings and wins 61 of 72 paired tasks against it, tying two and losing nine, so the macro advantage does not imply a uniform taskwise ordering. Appendix B.5 gives the per-task breakdown and the scope of each comparison.

Obs 4. Near-original performance at 25% gives way to consistent losses at 50%. The 25% RAZOR checkpoints remain close to or above the originals in macro score, whereas 50% removal is lossy on all four backbones. For example, GLM-4.7-Flash falls from 44.5 at 25% to 40.1 at 50%, against 44.8 unpruned. Qwen3.6-35B-A3B falls from 62.1 to 58.7, against 61.6. These small gains above the original macro scores do not establish that pruning improves the original models.

3.4 GENERATION BEHAVIOR (RQ3)

A task score does not describe the diversity, format, length, or termination of generated responses. We examine these properties on Qwen3.6-35B-A3B (Figure 3). Diversity compares all four criteria

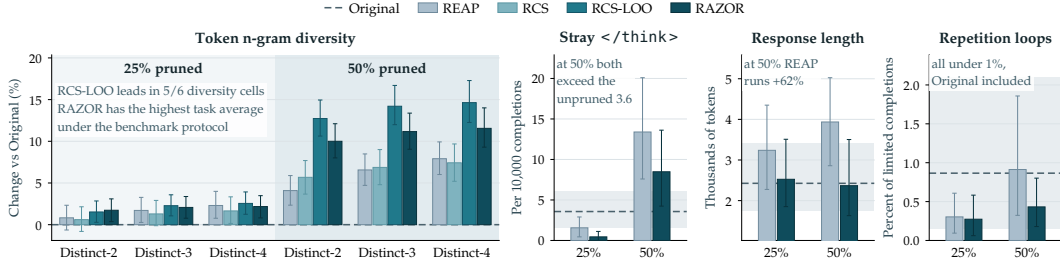


Figure 3: **Qwen3.6-35B-A3B response diversity and response form.** The left panel shows Distinct- n changes from Original. The right panels show completions containing `</think>` per 10,000 LiveCodeBench completions, median token length, and loop share among length-limited completions. Dashed lines mark Original, with 95% bands in the form panels. Whiskers show pointwise 95% bootstrap intervals.

by Distinct- n change from Original ($n = 2, 3, 4$) over 16 fixed responses per question, weighting AIME’26, HumanEval+, and LiveCodeBench equally. Response form compares REAP and RAZOR on LiveCodeBench over 256 shared sample positions per question (44,800 completions per variant). Sampling uses temperature 0.7, top- $p = 0.95$, top- $k = 20$, an 8,192-token cap, and thinking disabled, so these results probe response behavior rather than explicit reasoning traces. Diversity intervals use paired-question resampling and form intervals resample questions within each variant, both conditioning on the fixed generations (Appendices E.5 and E.6).

Obs 5. Higher diversity does not identify the best pruning criterion. RCS-LOO has the highest Distinct- n in five of the six combinations of removal budget and n -gram order, and exceeds REAP in all six by a median of 3.71 percentage points in relative change from Original. It does so despite RAZOR’s stronger macro task performance. The exception is Distinct-2 at 25%, where RAZOR changes by +1.73% from Original versus +1.53% for RCS-LOO. Changes are smaller at 25% than at 50%. In a separate four-task analysis, Qwen3.6-35B-A3B’s RCS-LOO–REAP Distinct-4 gap reverses under a 128-token prefix control that also changes question eligibility, so the reversal cannot be attributed to length alone (Appendix E.5). Higher diversity therefore establishes neither correctness nor a ranking robust to this joint control.

Obs 6. Closer response form does not imply preserved termination rates. At 50% removal, the rate of completions containing a stray `</think>` falls from REAP’s 13.4 to RAZOR’s 8.5 per 10,000 completions, whereas RCS-LOO reaches 68.5 (Table 10). Median length is also closer to Original, at 2.37k tokens for RAZOR versus 2.43k unpruned and 3.94k for REAP, as plotted in Figure 3. That table reports median character lengths for the same responses. Their proportional changes differ from those of median token lengths (Appendix E.6). These response-form comparisons draw on a different sample pool from the diversity result above and are not adjusted for length.

Length-limited finishes nevertheless reach 25.8% for RAZOR and 24.2% for REAP, against 16.0% unpruned. Few capped responses satisfy the high-repetition criterion. Among length-limited completions, loop shares under the rightmost panel’s criterion (fewer than one quarter of token 4-grams unique) are 0.27%/0.43% for RAZOR and 0.30%/0.91% for REAP at 25%/50% removal, all below 1% (Appendix E.6).

3.5 DISCUSSION (RQ4)

The checkpoint comparisons combine several scoring choices. We next vary individual components to assess their effects on predictive fidelity and expert selection.

3.5.1 SCORING COMPONENTS

The three criteria separate components of the local deletion counterfactual. RCS scores the routing-weighted consensus residual, RCS-LOO adds survivor renormalization, and RAZOR adds router-selected replacement. Table 6 compares them against REAP on matched GLM-4.7-Flash and Qwen3.6-35B-A3B checkpoints at both budgets, and the component contrasts of Appendix C.6 hold

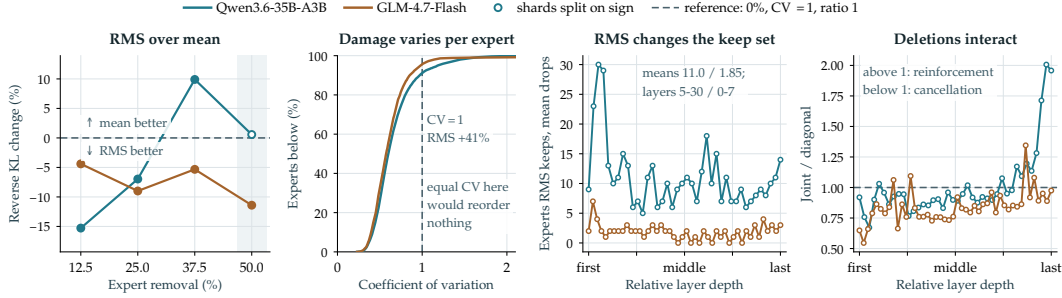


Figure 4: **RMS aggregation and expert selection for RAZOR.** Left to right, panels show relative KL change from mean (negative favors RMS), expert CV, RMS-only retained experts per layer, and deletion-set Gram ratios. Shading marks 50% removal, used in the last three panels.

the remaining choices fixed. The factorial study (Appendix C.5) comes from a separate evaluation collection and is therefore read on its own.

Obs 7. Consensus subtraction helps, while counterfactual refinements have mixed effects. Consensus subtraction lowers KL in all four model-budget settings when routing weights and mean aggregation are held fixed, though the margin is 0.2% in one of them, so the direction is consistent while the size is not always meaningful. This matched contrast isolates the reference change, unlike REAP-to-RCS, which also changes aggregation. Subsequent refinements are model-dependent. Under conditional RMS, adding the LOO factor lowers KL by 0.6%/5.6% on GLM-4.7-Flash but raises it by 7.1%/11.8% on Qwen3.6-35B-A3B at 25%/50% removal. Refill then improves on RCS-LOO in three of four settings, with GLM-4.7-Flash at 25% the exception (+0.5% KL). Thus a more complete single-deletion counterfactual need not produce a better jointly pruned checkpoint. The choice of RAZOR rests on the benchmark comparisons as well as these fidelity results, rather than local exactness alone.

3.5.2 EXPERT SELECTION

To relate aggregation to the retained experts, Figure 4 compares RMS-versus-mean KL across four budgets, per-expert coefficient of variation (CV), per-layer keep-set swaps, and deletion-set Gram ratios, with Table 7 placing RAZOR and RCS-LOO statistics side by side. Hollow markers in the KL panel indicate disagreement in sign across its four evaluation shards, and the latter three panels use 50% removal.

Obs 8. RMS changes retained sets, with model-dependent fidelity benefits. RMS yields lower KL than mean aggregation at all four budgets on GLM-4.7-Flash, but higher KL above 25% removal on Qwen3.6-35B-A3B. This differs from RCS-LOO, whose RMS-over-mean gains at 25% and 50% removal are small on Qwen3.6-35B-A3B (Appendix C.6). Variation in the RMS premium $\sqrt{1 + CV^2}$ reweights experts rather than uniformly rescaling them. At 50% removal, RMS retains an average of 11.0 experts per layer that mean would discard on Qwen3.6-35B-A3B and 1.85 on GLM-4.7-Flash, from retention budgets of 128 and 32, respectively. The per-layer ranges are 5–30 and 0–7. This establishes changed selection, not that the swaps cause benchmark gains. Appendix C.3 gives further aggregation contrasts.

Obs 9. Local diagnostics expose interactions that scalar rankings do not resolve. The Gram-ratio panel shows net reinforcement in four GLM-4.7-Flash layers for RAZOR, whereas RCS-LOO-selected sets show net cancellation in all 46 GLM-4.7-Flash layers and 29 of 40 Qwen3.6-35B-A3B layers (Appendix C.7.2). These interactions concern the fixed-support numerator, constrained by $\sum_i w_i(f_i - c) = 0$. The denominator and refill remain outside this probe. Single-deletion rankings also show no consistent advantage over REAP, despite favorable RCS-LOO comparisons in concentrated-routing bins (Appendices C.7.1 and C.7.3). Thus the probes expose interactions without establishing optimal joint selection or the cause of the benchmark gains.

4 RELATED WORK

MoE compression methods differ in the removal unit, the importance signal, and the selection procedure applied afterwards. RAZOR varies only the second, removing whole experts under a fixed layerwise budget without updating the learned expert or router weights. We compare it with methods that score whole experts independently, model relations among experts, or search over retained sets. Appendix F treats the families that change the removal unit or the surviving computation, including merging, router adaptation, budget allocation, and finer-grained compression.

Expert scoring. At the individual-expert level, pruning methods use routing, activation, or weight statistics (Muzio et al., 2024; Jaiswal et al., 2025; Liu et al., 2026a). REAP analyzes promoted substitution and survivor renormalization, but its pruning score averages the removed expert’s routing-weighted output norm over tokens selecting that expert (Lasby et al., 2026). A unified formulation separates the removed contribution from a rerouting residual and adopts a magnitude proxy, motivated by the cost of direct rerouting evaluation and the high overlap between proxy-based and exact-damage selections in its experiments (Liu et al., 2026b). Its score family varies routing frequency, gate weighting, and activation strength, recommending gate-free conditional-mean activation scores for task-agnostic pruning. Under our fixed-input routing assumptions, consensus residuals express the refill-aware perturbation using the routed outputs and one additional expert output per token. This avoids constructing a separate pruned mixture for every candidate deletion. Conditional RMS aggregation remains a separate design choice.

Expert relations. Independent scores can miss structure shared across experts. STUN and HC-SMoE cluster router behavior or expert outputs, while SHAPE, ConMoE, and MAESTRO use routing co-occurrence, parameter-space replaceability, or cross-layer transitions (Lee et al., 2025; Chen et al., 2025; Zhang, 2026; Yao et al., 2026; Goel et al., 2026). These relations support selection, merging, or path modeling, whereas RAZOR scores deletion with survivor renormalization and refill, without recovery.

Candidate-set search. Beyond relational ranking, other methods evaluate retained sets through reconstruction, joint pruning–merging, or layerwise and blockwise search (Lu et al., 2024; Liu et al., 2024; Yang et al., 2024). RAZOR performs no subset search. Its exactness for one fixed-input deletion does not solve joint pruning, because deletions interact and a refill candidate may itself be removed.

5 CONCLUSION

RAZOR uses consensus residuals to score expert replaceability under survivor renormalization and router refill. The resulting single-deletion identities support training-free pruning from forward computation. Across two backbones and two removal budgets, RAZOR achieves the highest nine-task macro average among the evaluated baselines and residual criteria, and lowers reverse KL relative to REAP in all four settings. It also achieves the highest macro task score among the residual criteria on two additional backbones. These results support including survivor and replacement outputs in expert scoring, while leaving the connection between local damage and joint pruning empirical. In particular, more complete local counterfactuals do not consistently improve fidelity, and residual criteria yield less stable selections than REAP in the Qwen3.6-35B-A3B calibration-budget study (Appendix D.1). Changes in diversity, formatting, and termination further show why task performance and response behavior require separate evaluation.

Limitations. Joint pruning can remove the refill candidate assumed by a single-deletion score. Interacting deletions and changes propagated across layers also remain outside the local counterfactual. The benchmark comparisons cover fixed checkpoints and report no task-level uncertainty. Physical pruning reduces stored parameters, but retaining the original top- k does not imply a proportional reduction in active computation. Scoring runtime, serving latency, energy use, and peak memory have not been measured.

AI USE STATEMENT

Generative AI tools assisted with the refinement of academic prose, focusing on clarity, coherence, and precision of expression. They also supported literature search and coding.

ETHICS STATEMENT

This work studies expert pruning of post-trained MoE language models, aiming to reduce model storage requirements while retaining their capabilities. Removing experts can also alter behaviors shaped by post-training, including safety-related responses, even when aggregate task performance is preserved. Agreement with the original model on safety-related inputs measures predictive fidelity, but does not establish safety or preservation of alignment. We therefore emphasize independent safety and alignment evaluation before deployment, particularly in safety-sensitive applications. The study uses previously released calibration data (Appendix B.1) and involves no human-subject research.

REPRODUCIBILITY STATEMENT

Section 2 and Appendix A provide the scoring criterion and derivations. Appendix A.4 gives the algorithm and implementation details. Appendix B documents calibration data, model configurations, evaluation protocols, score aggregation, and uncertainty estimates. Appendix B.5 states the evaluator settings and the scope of each cross-study comparison. The scoring and pruning implementation, the RAZORCAL calibration corpus, and the eight pruned checkpoints evaluated here are publicly available. Each checkpoint includes the retained expert indices. The public calibration release replaces potentially identifying strings with placeholders, so it need not reproduce the original expert selection exactly.

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Appendix A provides additional derivations and implementation details, and Appendix B specifies the data and evaluation protocols. Appendices C and D analyze scoring components, expert selection, and calibration sensitivity. Appendix E reports detailed fidelity and generation results, and Appendix F discusses related compression methods.

Scope of the reported estimates. All reported performance intervals condition on fixed checkpoints and exclude variability over calibration draws, pruning runs, and independent training runs. Matching a scoring rule across two studies does not imply that they share calibration inputs or retained expert sets, so fidelity and benchmark results are compared within a study rather than across studies.

A ADDITIONAL DERIVATIONS AND IMPLEMENTATION

We derive the geometry of consensus residuals, relate them to exact single-deletion damage, and examine interactions under joint removal. Appendix A.4 describes score accumulation and physical pruning.

A.1 RESIDUAL DECOMPOSITION

The consensus residual depends on the angle between an expert output and the mixture, which is the information that gates and output norms omit. Expanding it gives

$$\|f_i - c\|_2^2 = \|f_i\|_2^2 - 2\langle f_i, c \rangle + \|c\|_2^2, \quad (5)$$

$$\langle f_i, c \rangle = \|f_i\|_2 \|c\|_2 \cos \phi_i. \quad (6)$$

Here ϕ_i is the angle between nonzero f_i and c . If either vector is zero, their inner product is zero without defining an angle. At fixed norms, a positive inner product reduces the residual and a negative one increases it. Even without collinear or near-duplicate outputs, an aligned expert may therefore be replaceable, while one balancing the others may be costly to lose. The mixture still includes expert i , so a further self-inclusion correction is needed to obtain distance to the survivors. Because mixture norms, gates, and routed sets vary across calibration tokens, this geometric identity constrains each token’s damage without determining which retained set performs better.

Constructed example in Figure 1. Let u, v be orthonormal and let E_A, E_B, E_C have outputs $8u, 3u + 4v$, and $10u - 4v$, with equal initial weights. Their consensus is $c = 7u$, the routed-expert mixture at one layer, not the model’s final output. Residuals are $\|f_i - c\|_2$. Without refill, deleting E_A, E_B, E_C and renormalizing the survivors gives damage 0.5, $\sqrt{8} \approx 2.83$, and 2.5, respectively. With refill, the same initially unselected $E_D = 10u - 5v$ enters after every deletion, with survivor and replacement masses 1, 1, 0.8. For deletion of E_C , the bars show $-(f_C - c), 0.8(f_D - c)$, and their unnormalized sum. The v terms cancel, giving $\Delta c = -0.6u/2.8$ and damage $3/14 \approx 0.21$, versus 1.51 for deleting E_A and 3.66 for deleting E_B . Thus magnitude, fixed-support damage, and refill damage favor E_B, E_A, E_C , respectively.

In the panels, curves span output coordinates, with mixture curves using $2.5\times$ the expert vertical scale. Dotted curves isolate u components, and shading marks coordinate-wise differences. The construction illustrates the role of output geometry and refill rather than a measured checkpoint-level gain.

A.2 SURVIVOR-DISTANCE IDENTITY

For $w_i < 1$, starting from the fixed-support post-removal output gives

$$c^{-i} = \frac{c - w_i f_i}{1 - w_i}, \quad (7)$$

$$f_i - c^{-i} = f_i - \frac{c - w_i f_i}{1 - w_i} = \frac{f_i - c}{1 - w_i}. \quad (8)$$

Taking norms gives the survivor-distance identity used in the main text,

$$\|f_i - c\|_2 = (1 - w_i) \|f_i - c^{-i}\|_2, \quad (9)$$

and combining equation 9 with Proposition 3 yields

$$\|c - c^{-i}\|_2 = w_i \|f_i - c^{-i}\|_2, \quad (10)$$

which expresses damage as routing mass times distance to the survivors. The distance describes local functional replaceability at the same input, while the weight converts that distance into mixture change. Neither factor alone measures redundancy in the final retained model.

The factor $1/(1 - w_i)$ is what converts the observed consensus residual into this survivor distance. Omitting it yields $(1 - w_i)\delta_i^{\text{loo}}$ at each token, which is a different proxy rather than the same deletion damage under a rescaling, because w_i varies across tokens. The identity also makes the local comparison translation invariant. With inputs and gates fixed, adding a common vector g to all routed outputs shifts both mixtures by g and leaves deletion damage unchanged, although the output norms that magnitude scores rely on do change. This invariance holds for the local damage, not for model predictions.

For $\delta_i^{\text{loo}} > 0$, expanding the refill numerator gives

$$\frac{\delta_i}{\delta_i^{\text{loo}}} = \frac{1 - w_i}{1 - w_i + w_r} \sqrt{1 - 2\rho_i \cos \theta_i + \rho_i^2}, \quad \rho_i = \frac{w_r \|r_r\|_2}{w_i \|r_i\|_2}, \quad (11)$$

where θ_i is the angle between nonzero residuals. When the promoted weighted residual vanishes, the square-root factor is defined to be one. The denominator factor is at most one and strictly smaller when $w_r > 0$, whereas the numerator factor exceeds one exactly when $\rho_i > 0$ and $\rho_i > 2 \cos \theta_i$ both hold. The condition on ρ_i cannot be dropped, since $\rho_i = 0$ makes the factor exactly one. Orthogonal residuals thus already raise the numerator factor for any $\rho_i > 0$, yet routing mass, residual magnitudes, and alignment jointly decide whether refill increases the damage ratio itself. Appendix C.4 reports score ratios and partial geometric summaries, which measure the observed effect without decomposing it into these three factors.

A.3 EXACT PERTURBATION FOR A SELECTED SET

Joint removal couples the experts it deletes, so the exact single-deletion identities do not extend additively to a selected set. For a removed routed subset $\mathcal{A}$ with $W_{\mathcal{A}} < 1$, the survivor mixture is

$c^{-A} = (c - \sum_{j \in A} w_j f_j) / (1 - W_A)$. Subtracting it from c gives Eq. 4. All residuals $r_j = f_j - c$ share the original consensus, so the squared numerator expands as

$$\left\| \sum_{j \in A} w_j r_j \right\|_2^2 = \underbrace{\sum_{j \in A} w_j^2 \|r_j\|_2^2}_{\text{diagonal}} + \underbrace{\sum_{j \neq l \in A} w_j w_l \langle r_j, r_l \rangle}_{\text{interference}}. \quad (12)$$

Interference cross terms and the set-dependent denominator block an additive decomposition into independent removal costs. The diagonal term also sums experts within a token, whereas RMS aggregates tokens within an expert, so the expansion neither derives RMS nor gives an exact joint objective obtained by summing expert scores.

Under refill, let $\mathcal{R}$ contain the $|\mathcal{A}|$ highest-ranked unselected experts that remain in the retained pool. The numerator becomes $\| \sum_{j \in A} w_j r_j - \sum_{r \in \mathcal{R}} w_r r_r \|_2$ and the denominator becomes $1 - W_A + W_{\mathcal{R}}$, where $W_{\mathcal{R}} = \sum_{r \in \mathcal{R}} w_r$. Both now depend on the whole set, and the expansion acquires cross terms within $\mathcal{R}$ and between A and $\mathcal{R}$ on top of those in Eq. 12. Summing scalar single-expert scores captures none of these interactions, which is why our selection remains a ranking heuristic under a fixed budget.

A.4 IMPLEMENTATION AND COMPLEXITY

Algorithm 1 Chunked scoring and routing-aware physical pruning for the RCS family

Require: Model, calibration data $\mathcal{D}$ with valid-position masks, budgets B_ℓ , chunk size C , criterion $a \in \{\text{RCS}, \text{RCS-LOO}, \text{RCS-Refill}\}$, denominator clamp $\epsilon = 10^{-6}$

- 1: Initialize per-shard $q_{\ell i} \leftarrow 0$, $n_{\ell i} \leftarrow 0$ for learned-router layers
- 2: **for** learned-router layer input from an unpruned forward over $\mathcal{D}$ **do**
- 3: Obtain routed sets S_t and normalized selected weights w_t
- 4: **if** $a = \text{RCS-Refill}$ **then**
- 5: Obtain the rank- $(k+1)$ expert r_t and its pseudo-weight $w_r(t)$
- 6: **end if**
- 7: **for** token chunk X of size at most C **do**
- 8: Reconstruct routed outputs $f_i(t)$ and form $c_t = \sum_{i \in S_t} w_i(t) f_i(t)$ and $r_i(t) = f_i(t) - c_t$
- 9: **if** $a = \text{RCS}$ **then**
- 10: $d_i(t) \leftarrow w_i(t) \|r_i(t)\|_2$
- 11: **else if** $a = \text{RCS-LOO}$ **then**
- 12: $d_i(t) \leftarrow w_i(t) \|r_i(t)\|_2 / \max(1 - w_i(t), \epsilon)$
- 13: **else**
- 14: Reconstruct $f_{r_t}(t)$, set $r_{r_t}(t) = f_{r_t}(t) - c_t$
- 15: $d_i(t) \leftarrow \|w_i(t) r_i(t) - w_r(t) r_{r_t}(t)\|_2 / \max(1 - w_i(t) + w_r(t), \epsilon)$
- 16: **end if**
- 17: For valid routed pairs, accumulate $q_{\ell i} += d_i(t)^2$, $n_{\ell i} += 1$; discard chunk-local outputs
- 18: **end for**
- 19: **end for**
- 20: Sum q, n across disjoint shards; set $s_{\ell i} = \sqrt{q_{\ell i} / \max(n_{\ell i}, 1)}$
- 21: Retain the B_ℓ highest-scoring experts in each learned-router layer
- 22: For fixed-hash layers, select by table counts and remap to distinct surviving routes
- 23: **return** Model with consistently sliced expert tensors, router rows, and updated route indices

A.4.1 ONLINE STATISTICS AND BOUNDED WORKING BUFFERS

Algorithm 1 changes only the token-level quantity across the three criteria, which share chunking, conditional RMS aggregation, ranking, and physical pruning. The RCS and RCS-LOO branches reconstruct only the routed expert outputs, whereas RCS-Refill additionally reconstructs the rank- $(k+1)$ promoted expert. Selecting RCS-Refill with conditional RMS gives RAZOR. For each learned-router layer, the scoring pass obtains the routing context for the batch, partitions its token positions, and locally reconstructs the required expert outputs for each chunk. These outputs form the consensus and residuals before advancing to the next chunk. The cost is therefore additional local

computation alongside one original forward pass, rather than a separate end-to-end model evaluation for every candidate deletion.

Chunking bounds the working buffers. With T packed token positions, top- k routing, and expert-output width d , storing the routed outputs together with the promoted one requires $O(T(k+1)d)$ space. Restricting the working set to C positions replaces this with $O(C(k+1)d)$, plus $O(Cd)$ for the consensus. Only $q_i = \sum_t d_i(t)^2$, where $d_i(t)$ is whichever of the three token quantities the criterion selects, and the valid routed-token count n_i persist for RMS scoring, requiring $O(E)$ state per layer. Disjoint calibration shards accumulate these quantities independently. Summing them before taking $\sqrt{q_i / \max(n_i, 1)}$ gives the pooled conditional RMS, unlike averaging shard-level RMS scores. Unobserved experts receive zero score, and we remove the lowest $E - B$ scores without a secondary criterion for exact ties. These bounds cover only the storage controlled by chunking, excluding model weights and ordinary forward activations.

One implementation choice affects the computed numbers. Each denominator is clamped from below by 10^{-6} , giving $\max(1 - w_i + w_r, 10^{-6})$ for the refill score and $\max(1 - w_i, 10^{-6})$ for its fixed-support special case. Because the identities describe the unclamped quantity, an active clamp makes the computed score differ from exact local damage.

A.4.2 LAYER-WISE EXECUTION

Output chunking bounds activation buffers but leaves the memory occupied by model weights untouched. For models that exceed the available device capacity, a layer-wise schedule loads one decoder layer, propagates the current calibration subset through it, collects its statistics, and releases its weights before loading the next layer. This is what makes the largest backbones here scorable on a fixed device budget. The schedule preserves the score definition, although floating-point differences can affect ties or nearly tied rankings.

A.4.3 ARCHITECTURE-SPECIFIC ROUTING

The derivation requires normalized selected weights, not a particular router parameterization. We preserve each architecture’s native selection rule and score function, using softmax probabilities for Qwen3.6-35B-A3B, sigmoid scores for GLM-4.7-Flash, DeepSeek-V4-Flash-0731’s native score transformation, and sigmoid scores for Hy3. Where a correction bias determines selection, it is not added to the mixture weights. Selected unmodified scores are renormalized before evaluating the damage formula. The implementation retains the routed-output scale λ , although omitting this common positive factor from the displayed score leaves within-layer rankings unchanged.

DeepSeek-V4-Flash-0731 includes three fixed-hash layers whose expert identities come from a frozen token-to-expert table, so they fall outside learned-router saliency and need a separate rule. To retain B experts, we select the most frequent experts by counting occurrences over the entire table, without weighting by calibration-token occurrence. Each table row then preserves its surviving expert identities, and removed entries are replaced, in slot order, by the cosine-nearest surviving router-weight vector not already used in that row. With $B \geq k$, the result has k distinct routes, and reindexing follows the same survivor order as parameter slicing. The procedure repairs routing rather than merging expert weights or solving an assignment problem. The routing-stratified study shares this hash-layer selection across criteria, so its comparisons isolate learned-router saliency (Appendix B.5).

A.4.4 PHYSICAL PRUNING

We slice expert-indexed tensors and corresponding router rows with the same keep indices, leaving dense blocks and shared experts unchanged. Under a common retained width, the result is a smaller MoE checkpoint requiring neither runtime masks nor custom sparse kernels. For Qwen3.6-35B-A3B, reducing 256 to 128 routed experts per layer yields an approximately 19B-parameter model.

Removing experts reduces stored parameters, but preserving top- k does not proportionally reduce active expert computation. We do not measure serving latency, energy use, or end-to-end acceleration.

B DATA AND EVALUATION PROTOCOL

We use separate protocols for calibration, predictive fidelity, downstream performance, and response behavior. The calibration pool RAZORCAL scores experts (Appendix B.1), the eight-axis reference pool measures predictive fidelity (Appendix B.2), the nine-task suite measures downstream performance (Appendix B.5), and a separate repeated-generation collection measures response behavior (Appendix E.5). Only the first influences which experts are retained. The fidelity pool alone splits its eight axes into four that RAZORCAL covers and four that it does not, so calibration-covered and held-out fidelity can be reported separately throughout Appendix E. That ID/OOD distinction applies to the fidelity axes only. The benchmark suite and the repeated-generation collection are not partitioned this way, since they draw on public benchmarks rather than on axes defined relative to RAZORCAL. The calibration pool and the fidelity pool draw mainly on open Nemotron post-training datasets, whose development and reuse across the model family are described in the Nemotron 3 Super technical report (NVIDIA, 2026), together with a few non-Nemotron instruction sources named below. The nine benchmark tasks, and the generation analyses using those tasks, come from their own public releases (Appendix B.5). The specific dataset versions are listed below.

B.1 CALIBRATION DATA

RAZORCAL contains 2,048 chat-format examples across seven domains (Table 3), whose embedding spread and per-domain length coverage are shown in Figure 5. It is a calibration mixture assembled from existing instruction datasets, not a new independently collected corpus. Coding receives twice the allocation of each other domain to cover function completion, competitive programming, class-based solutions, software engineering, and tool use. Source-specific sampling applies quotas, length filters, and deduplication. Preprocessing removes dataset-specific harness prefixes where appropriate. The 32,000-character cap counts message content and serialized tool calls, excluding separately stored reasoning fields and tool schemas. Each checkpoint’s chat template renders the preserved conversation roles before tokenization.

Sources and composition. Coding includes 205 Python-algorithm examples from Dolci-Instruct-SFT, the instruction-tuning mixture accompanying Olmo 3 (Team Olmo et al., 2025). It adds 141 Python and 115 C++ examples from Nemotron-SFT-Competitive-Programming-v2, 40 agentless software-engineering examples from Nemotron-SFT-SWE-v2, and 11 general tool-use examples from Nemotron-SFT-OpenCode-v1. Nemotron-SFT-SWE-v2 covers code localization, repair, and test generation, whereas Nemotron-SFT-OpenCode-v1 covers agent interactions.

Mathematics contains 256 problem–solution examples from the high-reasoning AoPS subset of Nemotron-Math-v2, including 96 selected for competition-style problems. This source pairs problems from Art of Problem Solving with model-generated solutions. High denotes the teacher’s reasoning mode, not problem difficulty. Science combines 176 chemistry reasoning questions and 80 multiple-choice questions from Nemotron-Science-v1. Chinese-STEM uses translated mathematics, code, and STEM examples from Nemotron-SFT-Multilingual-v1 (86/85/85), which translates prompts and final answers rather than all reasoning traces. Instruction following draws 256 examples from a Dolci constraint-following pool (Team Olmo et al., 2025). Tool calling uses 256 multi-turn agent–tool conversations from Nemotron-Agent-v1. The world-knowledge allocation contains 256 OpenHermes-2.5 examples (Teknum, 2023), drawn from the cleaned, non-thinking subset in SmolTalk2 (Hugging Face Smol Models Research, 2025). OpenHermes is a general-purpose synthetic instruction corpus rather than a dedicated knowledge test, and all quotas balance source examples rather than token counts.

Scored positions and coverage. For the diagnostic collection, calibration examples are rendered with thinking disabled, separately supplied reasoning fields omitted, and packed into 4,096-token rows. Calibration statistics include all nonpadding conversation positions, including prompts and role headers, unlike the assistant-only fidelity targets in Appendix B.2. Packing fills each row in order and stops at the row boundary, so an example that crosses it is truncated and its tail is discarded rather than continued into the next row. Long examples therefore contribute only their leading tokens, and the collection stops once the token budget is filled, which is why the counts below cover fewer than the 2,048 pool examples. Qwen3.6-35B-A3B scores 785,703 valid tokens from 921 source examples, while GLM-4.7-Flash scores 785,632 tokens from 992. We call these the *diagnostic calibration*

Table 3: RAZORCAL domain allocations and calibration sources. Counts denote source examples, not tokens.

Domain	Samples	Source
Coding	512	Dolci-Instruct-SFT and Nemotron competitive programming, SWE, and OpenCode
Mathematics	256	Nemotron-Math-v2, high-reasoning AoPS subset
Science	256	Nemotron-Science-v1
Chinese-STEM	256	Nemotron-SFT-Multilingual-v1
Instruction following	256	Dolci constraint-following mixture
Tool calling	256	Nemotron-Agentive-v1
World knowledge	256	OpenHermes-2.5

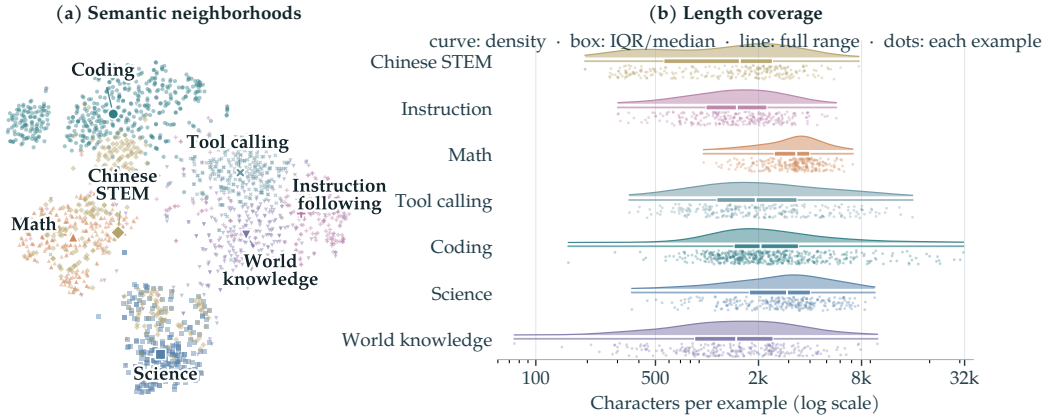


Figure 5: **Overview of the RAZORCAL calibration pool.** (a) Cosine t-SNE projection of Qwen3-Embedding-8B representations of the rendered conversations, with color and marker shape encoding domain and labels marking domain medians. (b) Per-domain character-length coverage over the full observed range, counting message content only. The curve is a kernel-density estimate in log-character space, the bar is the interquartile range with the median marked, the thin line spans minimum to maximum, and each dot is one of the 2,048 examples. Domain counts are given in Table 3.

collections below. They provide the score-distribution and Gram-matrix statistics, as well as the saliency scores used in the single-expert ablations in Appendix C. The ablation damage is measured on separate evaluation inputs. Because the two tokenizers give different token counts and source coverage at the same row capacity, these statistics are compared within a backbone rather than across backbones.

B.2 FIDELITY EVALUATION DATA

Reverse KL and reference-token perplexity use a pool of 8,192 conversations, with 1,024 examples on each of eight task axes. These compare predictions on shared reference responses rather than scoring benchmark answers. Four axes cover calibration task types.

Mathematics. Mathematical problem-solving responses from the `medium` split of Nemotron-Math-v2. Medium denotes the teacher’s reasoning mode, not problem difficulty.

Code. Python solutions from the `competitive_coding_python` split of Nemotron-SFT-Competitive-Programming-v2.

Instruction following. Constraint-following conversations from the `chat_if` split of Nemotron-Instruction-Following-Chat-v1, selected by its instruction-following capability label.

Tools. Agent-tool interaction trajectories from the `tool_calling` split of Nemotron-Agentive-v1.

The other four axes hold out task types.

Chat. Examples carrying the chat capability label in the same Chat-v1 `chat_if` split.

Creative writing. Examples selected by prompt keywords from the `reasoning_off` split of Nemotron-SFT-Instruction-Following-Chat-v2. Creative writing is our selection, not an official split.

Safety. Reference safety-aligned responses from the `train` split of Nemotron-SFT-Safety-v1.

SQL. Examples from the `text_to_sql` split of the competitive-programming source, separate from its Python subset, pairing natural-language tasks and database schemas with SQL. ID/OOD denotes calibration task-type coverage, not disjoint source datasets or a formal distribution shift, and both groups contain synthetic data.

Each axis retains 1,024 examples that fit the row capacity and carry enough assistant content to score. Conversations end on an assistant response, and plain-text assistant turns may be truncated while structured tool calls remain intact.

Evaluation packs examples into 4,096-token rows and scores assistant positions, excluding role headers and turn terminators. Chat rendering disables thinking and omits separately supplied reasoning fields, so fidelity is measured on fixed reference prefixes rather than on generated reasoning trajectories. The routing study uses six batches per axis (Appendix E.3), and its evaluated batches form a subset of this pool. The fidelity pool has no exact-message overlap with calibration, although near-duplicates and benchmark contamination have not been ruled out.

B.3 EVALUATION CONFIGURATION

Table 4 summarizes the four main backbones. Qwen3.6-35B-A3B uses the hybrid attention–MoE architecture described for Qwen3.5 (Qwen Team, 2026). The GLM and DeepSeek model families are described in their technical reports (GLM-4.5 Team, 2025; DeepSeek-AI, 2026). The configurations below refer to the evaluated checkpoints. Hy3 is the public release (Tencent Hy Team, 2026). The architecture and routing settings describe the language-model decoder. Layer counts exclude auxiliary prediction modules. GLM-4.7-Flash and Hy3 each begin with one dense feed-forward layer. All 43 DeepSeek-V4-Flash-0731 layers are MoE layers, of which three use fixed-hash routing and the other 40 use learned-router selection. Its score transformation is $\sqrt{\text{softplus}(z)}$. The nominal parameter counts exclude auxiliary MTP and DSpark modules.

Both removal ratios retain the original top- k and shared experts. Calibration uses the 2,048-example source pool, while the scored coverage in Appendix B.1 refers specifically to the Qwen3.6-35B-A3B and GLM-4.7-Flash diagnostic collection. For downstream benchmarks, we use temperature 0, except for SWE-bench Verified, where we use 0.7. AIME’26 reports avg@8, and the other eight benchmarks report avg@3. For each benchmark, sampling parameters are shared across all four backbones. We set the maximum input length to 224K tokens for GLM-4.7-Flash and Qwen3.6-35B-A3B, and 128K tokens for DeepSeek-V4-Flash-0731 and Hy3. All four use a 32K-token output limit. These are evaluation limits, not the models’ maximum context capacities. The Qwen3.6-35B-A3B and GLM-4.7-Flash benchmark settings enable thinking. Fidelity and response diversity use their own protocols, with generation analysis using temperature 0.7. Cross-study comparability is discussed in Appendix B.5.

Comparisons among the residual criteria match calibration inputs and layerwise budgets within each study. Benchmark and diagnostic protocols use separate checkpoints and calibration draws, as specified in Appendix B.5.

B.4 METRIC DEFINITIONS

Let p_t and q_t be the unpruned and pruned next-token distributions on the same reference prefix. Reverse KL is

$$D_{\text{KL}}(q_t \| p_t) = \sum_v q_t(v) \log \frac{q_t(v)}{p_t(v)}. \quad (13)$$

The main fidelity sweep weights assistant tokens within each evaluation axis and balances the eight axes equally.

Table 4: Architectures and pruning settings of the four main backbones. Parameter counts are nominal backbone sizes, and expert counts are per MoE layer.

	Qwen3.6-35B-A3B	GLM-4.7-Flash	DeepSeek-V4-Flash-0731	Hy3
<i>Architecture</i>				
Total parameters	35B	30B	284B	295B
Active parameters per token	~3B	~3B	~13B	~21B
Decoder layers	40	47	43	80
MoE layers	40	46	43	79
Routed experts	256	64	256	192
Expert intermediate width	512	1,536	2,048	1,536
Shared experts	1	1	1	1
<i>Routing</i>				
Top- k	8	4	6	8
Router scores	softmax	sigmoid	sqrtsoftplus	sigmoid
Routed scale λ	1.0	1.8	1.5	2.826
Fixed-hash layers	0	0	3	0
<i>Pruning</i>				
Pruned fractions		25% and 50%		
Retained routed experts (25%)	192	48	192	144
Retained routed experts (50%)	128	32	128	96

For reference token y_t , define $\ell_p(t) = -\log p_t(y_t)$ and $\ell_q(t) = -\log q_t(y_t)$. Averaging these losses under the specified evaluation weights gives reference-token NLL and

$$\begin{aligned} \text{PPL}_p &= \exp(\text{NLL}_p), & \text{PPL}_q &= \exp(\text{NLL}_q), \\ \Delta\text{NLL} &= \text{NLL}_q - \text{NLL}_p, & \frac{\text{PPL}_q}{\text{PPL}_p} - 1 &= \exp(\Delta\text{NLL}) - 1. \end{aligned} \quad (14)$$

The final quantity is excess perplexity. These likelihood metrics use observed reference tokens, not samples from p_t or q_t . Negative ΔNLL indicates higher reference-token likelihood under the pruned model, not necessarily better downstream capability.

For normalized selected routing weights and $k > 1$, routing entropy is

$$H_r(x) = -\frac{\sum_{j \in S(x)} w_j(x) \log w_j(x)}{\log k}. \quad (15)$$

Smaller H_r means more concentrated routing. Predictive entropy instead summarizes the vocabulary distribution,

$$H(p_t) = -\sum_v p_t(v) \log p_t(v), \quad \Delta H_t = H(q_t) - H(p_t). \quad (16)$$

Absolute entropy drift averages predictive entropies within routing cells and across valid layers, takes the absolute pruned–original difference in each batch–decile cell, and then averages these differences. Taking the absolute value after cell aggregation distinguishes this measure from the mean per-token $|\Delta H_t|$. Predictive entropy also differs from reference-token NLL, so its exponential is not the PPL reported here. Appendix E specifies the sampling units and uncertainty estimates for each output analysis.

B.5 DOWNSTREAM BENCHMARK PROTOCOL AND SCOPE

The benchmark evaluates four backbones at two removal ratios each. Table 1 covers GLM-4.7-Flash and Qwen3.6-35B-A3B, and Table 2 covers DeepSeek-V4-Flash-0731 and Hy3. The full method comparison includes Frequency, EAN, REAP, and all three residual scores on GLM-4.7-Flash and Qwen3.6-35B-A3B. RCS, RCS-LOO, and RAZOR use conditional RMS. DeepSeek-V4-Flash-0731 and Hy3 compare the three residual scores against their unpruned references only, so they enter the RCS-LOO-relative and unpruned-relative benchmark comparisons rather than the REAP-relative ranges and win counts.

Baseline scores. For $n_i = |\mathcal{D}_i| > 0$ routed calibration tokens, the baseline scores are, up to common positive layer scales that do not affect ranking,

$$s_i^{\text{Frequency}} = n_i, \quad s_i^{\text{EAN}} = \frac{1}{n_i} \sum_{x \in \mathcal{D}_i} \|f_i(x)\|_2, \quad s_i^{\text{REAP}} = \frac{1}{n_i} \sum_{x \in \mathcal{D}_i} w_i(x) \|f_i(x)\|_2. \quad (17)$$

Frequency counts hard routing selections, as in frequency-based pruning (Muzio et al., 2024; Jaiswal et al., 2025). EAN is our conditional-mean adaptation of activation-norm pruning (Jaiswal et al., 2025; Lasby et al., 2026). We divide by each expert’s routed-token count, whereas the accumulated-norm EAN in REAP’s baseline definition does not. This expert-specific normalization is not a common rescaling and can change the ranking. REAP follows the routed-token conditional mean in its original definition (Lasby et al., 2026). The conditional-mean EAN score is algebraically identical to MAN in Liu et al. (2026b). We retain the EAN label in the tables. All three scores are computed without recovery training, and zero-observation experts receive zero score.

Mathematics and instruction following. The nine tasks cover six capability categories. Each is graded on its final answer or artifact rather than on the reasoning that produced it, so a task that requires many dependent steps is still scored by whether the endpoint is correct. AIME’26 uses the 2026 American Invitational Mathematics Examination (Mathematical Association of America, 2026) to assess competition mathematics. IFEval (Zhou et al., 2023) checks compliance with programmatically verifiable instructions. IFBench (Pyatkin et al., 2025) extends this evaluation to additional verifiable constraints, testing instruction-following generalization beyond IFEval.

Knowledge, tools, and long context. SuperGPQA (Du et al., 2025) evaluates graduate-level knowledge across 285 disciplines. The Berkeley Function Calling Leaderboard, version 4 (BFCL v4, Patil et al., 2025; Mao et al., 2025), evaluates function calling and agentic tool use. LongBench v2 (Bai et al., 2025) tests comprehension and reasoning over long contexts.

Coding tasks and versions. The three coding tasks cover function-level generation, competition programming, and repository-level issue resolution. HumanEval+ (Liu et al., 2023) uses EvalPlus’s expanded tests to assess the functional correctness of generated Python functions. LiveCodeBench (Jain et al., 2025) draws from time-indexed programming competitions. We evaluate LiveCodeBench 2026, abbreviated LCB’26 in the tables, using the latest version in our evaluation setup. This evaluation is distinct from the 175-question subset used for the generation diagnostics (Appendix E.5). SWE-bench Verified (Jimenez et al., 2024), abbreviated SWE, is the 500-issue subset of SWE-bench screened by software developers for well-specified issues and appropriately scoped tests.¹ Scores report the percentage of resolved issues averaged over three evaluations.

Score aggregation. Instruction-following and coding averages give equal weight to their two and three tasks, respectively, while Overall weights all nine equally. Averages and paired macro gaps are computed from task scores rather than rounded displayed means. The task scores yield a 33/0/3 win/tie/loss count for RCS-LOO against REAP and 36/0/0 for RAZOR across GLM-4.7-Flash and Qwen3.6-35B-A3B at two budgets. Two properties qualify these counts. Both sets of 36 comparisons reuse one task suite rather than replicate it, and neither extends to the two backbones without a REAP benchmark run. The benchmark comparisons report no task-level uncertainty. AIME’26 reports avg@8 and the other eight benchmarks report avg@3. The three residual criteria and REAP use matched RAZORCAL inputs, whereas REAP* is an external GLM-4.7-Flash checkpoint calibrated with 24,576 samples.

Additional task contrasts. On Hy3, RAZOR scores 63.4 and 59.5 at 25% and 50% removal, respectively, versus 63.7 unpruned. On DeepSeek-V4-Flash-0731, the corresponding scores are 61.0 and 55.6 versus 59.2 unpruned (Table 2). The macro ordering is neither uniform across tasks nor monotonic across scoring refinements. At 25% on Qwen3.6-35B-A3B, RCS slightly exceeds RCS-LOO in macro average (61.0 versus 60.8). At 50% on GLM-4.7-Flash, EAN exceeds RAZOR on LCB’26 (33.9 versus 29.8).

For RCS-LOO versus REAP, both backbones carrying that comparison show larger macro gaps at 50% than at 25% (0.80 to 3.01 on GLM-4.7-Flash and 1.02 to 1.96 on Qwen3.6-35B-A3B), leaving

¹https://huggingface.co/datasets/princeton-nlp/SWE-bench_Verified

open whether the widening is general. On DeepSeek-V4-Flash-0731 at 25%, RCS-LOO gains on SWE (12.4 to 18.8) but loses on HE+ (91.5 to 89.6) relative to RCS, the nearest measured criterion on that backbone. At 50%, SWE suffers the largest proportional loss from the original for RCS-LOO on both Hy3 and DeepSeek-V4-Flash-0731. Thus even favorable macro comparisons coexist with capability-specific trade-offs.

Comparability. Benchmark and diagnostic results are compared within a study rather than across studies, since the two use separate checkpoints and calibration draws even when they share a scoring criterion. On DeepSeek-V4-Flash-0731, where the routing study matches fixed-hash selections across criteria, its benchmark gap therefore reflects the full pruned model rather than learned-router saliency alone. The LiveCodeBench comparison uses the same release across methods. Comparisons with external results require matching the task release and evaluation protocol. Input and output limits and shared sampling settings appear in Appendix B.3.

C SCORING PROXIES AND EXPERT SELECTION

A scoring rule makes two choices that the main text reports jointly, namely which quantity to measure at a routed token and how to reduce that quantity across tokens. This appendix separates them. Appendices C.2 through C.5 compare these choices and their interactions. Conditional RMS yields lower KL than the conditional mean for most tested token quantities, while the benefit of the residual reference depends on aggregation and backbone. Appendix C.7 then asks how the scoring choices change the retained set itself, through three probes that measure selection consequences rather than score values.

The analyses below use distinct evaluation collections. Comparisons are interpreted within each collection, except for the explicitly identified mean-refill contrasts in Appendix C.5, which are not fully matched.

C.1 SCORING CRITERIA AND BUDGET COMPARISONS

Table 5 lists the three consensus-residual scoring rules and the magnitude baseline REAP-RMS, all with conditional RMS, alongside their experimental coverage.

Each criterion fixes a token quantity, and every criterion here conditions its aggregation on tokens routed to the expert. RCS uses the routing-weighted consensus residual, RCS-LOO adds survivor renormalization, and RCS-Refill additionally models the promoted expert. The token quantities of RCS-LOO and RCS-Refill are exact under their respective deletion assumptions, yet the fixed-support identity fixes only the combination of consensus residual and LOO factor. It leaves both the aggregation rule and the performance of the selected checkpoint open, so the three rules need not form a performance hierarchy.

Magnitude and residual criteria can be compared under the same aggregation. REAP-RMS uses the output norm and differs from REAP solely in replacing the conditional mean with RMS, isolating aggregation from the reference change. Our EAN adaptation averages $\|f_i\|_2$ and REAP averages $w_i\|f_i\|_2$, with source definitions and the EAN normalization difference in Appendix B.5. The REAP-RMS results belong to the factorial collection of Appendix C.5, whose Table 9 crosses f_i against $f_i - c$ and w_i against $w_i/(1 - w_i)$ over the available sum, mean, and RMS reductions, with refill damage as an additional quantity.

Table 6 compares REAP, RCS, RCS-LOO, and RAZOR under the same checkpoint-evaluation protocol. Moving from REAP to RCS lowers KL on both models, but changes both the reference and mean-to-RMS aggregation, so it does not isolate consensus subtraction. The matched mean-aggregation contrast appears in Appendix C.6. Adding the LOO denominator raises KL on Qwen3.6-35B-A3B, while adding refill lowers it in three of four settings. These estimates pool tokens within each axis and weight the eight axes equally. The Δ vs REAP column gives RCS-LOO’s relative change from REAP, while refill gain gives RAZOR’s relative reduction from RCS-LOO. Negative values favor RCS-LOO in both columns, and percentages use unrounded KL.

Table 5: Scoring criteria and experimental coverage.

Instance	Token quantity	Evidence in this paper
RCS-Refill (RAZOR)	$\frac{\ w_i r_i - w_r r_r\ _2}{1 - w_i + w_r}$	Nine-task benchmarks on all four backbones, with baselines on two (Tables 1, 2). Reverse KL (Figure 2, Table 6), score distributions (Table 7), and refill mechanism (Table 8). Mean-RMS reduction and factorial comparisons (Figure 6, Table 9). Per-axis, routing-stratified and PPL fidelity, selection probes, and Qwen3.6-35B-A3B response behavior (Table 10).
RCS-LOO	$\frac{w_i \ r_i\ _2}{1 - w_i}$	Nine-task benchmarks on all four backbones, with baselines on two (Tables 1, 2). Output-fidelity and diversity figures, response behavior, reverse KL, score distributions, and calibration-budget and corpus diagnostics.
RCS	$w_i \ r_i\ _2$	Nine-task benchmarks on all four backbones, with baselines on two (Tables 1, 2). Reverse KL (Figure 2, Table 6) and the reduction axis (Figure 6).
REAP-RMS	$w_i \ f_i\ _2$	Reverse KL in the separate factorial collection (Appendix C.5). No benchmarks.

Table 6: Reverse KL across scoring rules in nats (lower is better).

Removal	REAP	RCS	RCS-LOO	RAZOR	Δ vs REAP	Refill gain
<i>Qwen3.6-35B-A3B</i>						
25%	0.07461	0.06669	0.07141	0.06872	−4.3%	+3.8%
50%	0.45654	0.40337	0.45108	0.44397	−1.2%	+1.6%
<i>GLM-4.7-Flash</i>						
25%	0.22850	0.20481	0.20351	0.20459	−10.9%	−0.5%
50%	1.01690	0.92310	0.87103	0.85457	−14.3%	+1.9%

C.2 RMS AGGREGATION AND SCORE DISTRIBUTIONS

For routed calibration tokens $\mathcal{D}_i = \{x \in \mathcal{D} : i \in S(x)\}$, RAZOR uses the conditional root mean square of refill damage,

$$s_i = \sqrt{\frac{1}{|\mathcal{D}_i|} \sum_{x \in \mathcal{D}_i} \left(\frac{\|w_i(x)r_i(x) - w_r(x)r_r(x)\|_2}{1 - w_i(x) + w_r(x)} \right)^2}. \quad (18)$$

This definition applies when $|\mathcal{D}_i| > 0$, with unseen experts handled as specified in Appendix A.4. The token damage follows from Proposition 2, but conditioning and RMS are aggregation choices, not consequences of that identity. The same aggregation is applied to $w_i \|r_i\|_2$ for RCS and δ_i^{loo} for RCS-LOO.

RMS differs from the conditional mean by an explicit variance term, which is what gives larger observed shifts more weight. With empirical conditional mean μ_i and population standard deviation σ_i of token-level damage over $\mathcal{D}_i$,

$$s_i = \sqrt{\mu_i^2 + \sigma_i^2} = \mu_i \sqrt{1 + \text{CV}_i^2}, \quad \text{CV}_i = \sigma_i / \mu_i \quad (\mu_i > 0). \quad (19)$$

For $\mu_i < \mu_j$, RMS ranks i above j exactly when $\sigma_i^2 - \sigma_j^2 > \mu_j^2 - \mu_i^2$. Such reversals change the retained set only across the budget boundary. This preference for variable damage motivates RMS as an empirical design choice rather than a guarantee of better pruning. Appendix C.6 tests that choice by comparing mean and RMS while holding the token quantity fixed.

For the score-distribution analysis, we additionally accumulate the first moment of token-level damage. Together with the routed-token count and second moment used for RMS scoring, this yields each expert’s mean and standard deviation with $O(E)$ additional state per layer. We collect

these statistics for both fixed-support and refill damage. Table 7 compares RCS-LOO and RAZOR. Equal-expert quantiles cover all observed experts without a count threshold, comprising 10,240 on Qwen3.6-35B-A3B and 2,944 on GLM-4.7-Flash, and the CV panel of Figure 4 uses the same experts. Within-layer dispersion is $\text{median}_\ell[\text{std}_i(\text{CV}_{\ell i}) / \text{mean}_i(\text{CV}_{\ell i})]$, using population standard deviations.

What changes a ranking is the premium’s variation across experts rather than its absolute size. The median fixed-support premium is 1.189 on Qwen3.6-35B-A3B and 1.155 on GLM-4.7-Flash. We quantify the resulting boundary swaps by comparing direct top- B mean and RMS rankings on the same experts and averaging swap counts equally across layers, with each entry reporting the RMS-only retained count over the retained-set size. Qwen3.6-35B-A3B has more exchanges at both budgets. Under refill, the median premium is 1.166 on Qwen3.6-35B-A3B and 1.153 on GLM-4.7-Flash, and the layer-averaged swap counts differ from the fixed-support averages by less than one expert. The two counterfactuals therefore show similar relative variability and similar mean–RMS selection differences, which is compatible with unequal damage magnitudes and unequal second moments between them.

One boundary crossing shows the mechanism concretely. We take Qwen3.6-35B-A3B layer 0, the first layer with a mean–RMS disagreement at 25% removal in the 4,096-token-row diagnostic, and within it the opposite-status pair with the closest conditional means, selected without reference to downstream outcomes. Expert 208 has mean/CV/RMS 0.03323/0.741/0.04136 over 32,041 routed tokens, against 0.03359/0.637/0.03982 over 23,766 tokens for expert 38. Their mean-to-RMS ranks move from 194 to 190 and from 189 to 201, respectively, crossing the 192-expert retention boundary. Higher variability thus flips selection across a conditional-mean margin of 0.00036, for one pair whose utility and presence in the benchmark checkpoints are not established here.

C.3 CONDITIONAL RMS, MEAN, AND CORPUS-SUM REDUCTIONS

The previous subsection compared conditional RMS against the conditional mean, which isolates the second moment. A second and independent choice remains, namely whether to condition on expert use at all, and the two are easily conflated because both appear to make RMS “weight large damage more.” Separating them requires a third reduction that drops the conditioning. Let g_i be any non-negative token quantity with conditional RMS s_i , let $\pi_i = |\mathcal{D}_i|/|\mathcal{D}|$, and set g_i to zero on tokens not routed to i . The empirical average squared damage over *all* calibration tokens is

$$\frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \mathbf{1}\{i \in S(x)\} g_i(x)^2 = \pi_i s_i^2. \quad (20)$$

An all-token RMS criterion would therefore rank $\sqrt{\pi_i} s_i$, which differs from ranking s_i by the exposure factor $\sqrt{\pi_i}$. Ranking s_i measures severity conditional on use, so an infrequent but severe deletion is not discounted for being rare. This score differs from the summed squared token quantity over all calibration tokens. Neither aggregation guarantees preservation of capabilities absent from the calibration data.

To isolate aggregation, we fix the RCS token quantity $g_i(t) = w_i(t) \|f_i(t) - c(t)\|_2$ and compare three reductions of the observed damage at four pruning budgets,

$$\begin{aligned} \text{conditional RMS} & \sqrt{\frac{1}{n_i} \sum_{t \in \mathcal{D}_i} g_i(t)^2}, & \text{conditional mean} & \frac{1}{n_i} \sum_{t \in \mathcal{D}_i} g_i(t), \\ \text{corpus sum} & \sum_{t \in \mathcal{D}} g_i(t)^2 = |\mathcal{D}| \pi_i s_i^2, \end{aligned} \quad (21)$$

where $\mathcal{D}_i$ contains tokens routed to expert i , $n_i = |\mathcal{D}_i|$, and $g_i(t) = 0$ elsewhere. The corpus sum differs from the all-token mean squared damage in Eq. 20 only by the common factor $|\mathcal{D}|$, so they induce the same ranking. We exclude selection frequency n_i because it discards g_i entirely. Including it would conflate a change in token quantity with a change in aggregation. Table 1 compares frequency as a criterion in its own right.

For the RCS quantity, reverse KL follows the strict ordering $\text{RMS} < \text{mean} < \text{sum}$ in all eight model–budget settings (Figure 6). Both choices thus point the same way under this protocol, as conditioning on use beats weighting squared severity by exposure, and squaring within the conditional reduction beats not squaring. The all-token quantity measures summed squared residual scores across calibration tokens. It remains a proxy and does not equal the error of jointly pruning the selected set.

Table 7: Damage variability and mean–RMS selection differences for RCS-LOO and RAZOR.

	Qwen3.6-35B-A3B		GLM-4.7-Flash	
	RCS-LOO	RAZOR	RCS-LOO	RAZOR
<i>Sample</i>				
Scored tokens	785,703		785,632	
<i>Coefficient of variation</i>				
CV median	0.643	0.599	0.577	0.574
CV p_5 / p_{95}	0.370 / 1.255	0.373 / 1.184	0.350 / 1.002	0.362 / 0.978
Within-layer dispersion of CV	0.317	0.305	0.221	0.211
<i>RMS premium</i> $\sqrt{1 + CV^2}$				
Median	1.189	1.166	1.155	1.153
p_{95}	1.605	1.550	1.416	1.398
<i>Boundary swaps</i>				
25% removal	7.95 / 192	7.53 / 192	1.39 / 48	1.33 / 48
50% removal	10.80 / 128	11.00 / 128	1.70 / 32	1.85 / 32

How much the reduction matters is easier to appreciate against the spread among criteria, which the main text reports as the headline comparison. Between the extreme reductions, KL spans $2.4\times$ and $1.7\times$ on Qwen3.6-35B-A3B and $1.15\times$ and $1.05\times$ on GLM-4.7-Flash at 25% and 50% removal. The corresponding spans among REAP, RCS, and RCS-LOO in Table 6 are $1.12\times$, $1.13\times$, $1.12\times$, and $1.17\times$. The reduction span is thus larger on Qwen3.6-35B-A3B, comparable on GLM-4.7-Flash at 25%, and smaller there at 50%. These spans describe different sets of tested configurations. Because the REAP–RCS–RCS-LOO comparison changes aggregation as well as token quantity, they do not establish which factor contributes more. The factorial grid in Appendix C.5 separates these choices.

The ordering largely carries over to the two exact deletion quantities. These comparisons use equal layerwise budgets on the same four shards as Table 6, applying the two conditional reductions to the leave-one-out and refill residuals. For refill damage, RMS lowers reverse KL relative to the conditional mean at all four GLM-4.7-Flash budgets, by 4.4%–11.4%, and at 12.5% and 25% on Qwen3.6-35B-A3B, while raising it at 37.5% and 50% on Qwen3.6-35B-A3B (0.20744 versus 0.18874, and 0.44397 versus 0.44143), for six of eight settings improved. Applied to the leave-one-out residual, RMS lowers KL in seven of eight settings, failing only at 37.5% on Qwen3.6-35B-A3B (0.20504 versus 0.19196). RMS therefore improves KL in most settings for both deletion quantities, with exceptions on Qwen3.6-35B-A3B at higher removal budgets.

C.4 WHAT THE REFILL TERM CHANGES

Refill improves macro task performance across the four evaluated backbones. To examine its local effect, Table 8 compares fixed-support and refill damage on the two diagnostic backbones using the calibration statistics of Table 7. Refill changes score scale in opposite directions while preserving high token-profile similarity for most experts. Router averages are token-weighted over routed events, whereas score ratios and token-axis cosines use unweighted expert quantiles. The routed-event-weighted mean $w_i = 1/k$ is a renormalization consistency check, since each token’s k selected weights sum to one.

Score scale and residual geometry. The median refill-to-fixed-support score ratio corresponds to a 2.81% increase on Qwen3.6-35B-A3B and a 2.56% decrease on GLM-4.7-Flash. Refill raises scores for 85.2% of Qwen3.6-35B-A3B experts and 14.3% of GLM-4.7-Flash experts. Substituting the reported mean routing weights into $(1 - w_i) / (1 - w_i + w_r)$ gives 0.922 and 0.850, respectively. These values illustrate the denominator’s attenuation at average weights. They are neither token-averaged factors nor a decomposition of the median score ratio.

For the reported residual-cosine statistics, the denominator $\|r_i\|_2 \|r_r\|_2$ is clamped below at 10^{-9} . The statistic is zero when either residual vanishes. The mean residual cosines are 0.108 on Qwen3.6-35B-A3B and 0.103 on GLM-4.7-Flash. By Eq. 11, such a low positive cosine is compatible with either an increase or a decrease in the numerator, depending on the relative residual magnitude ρ_i .

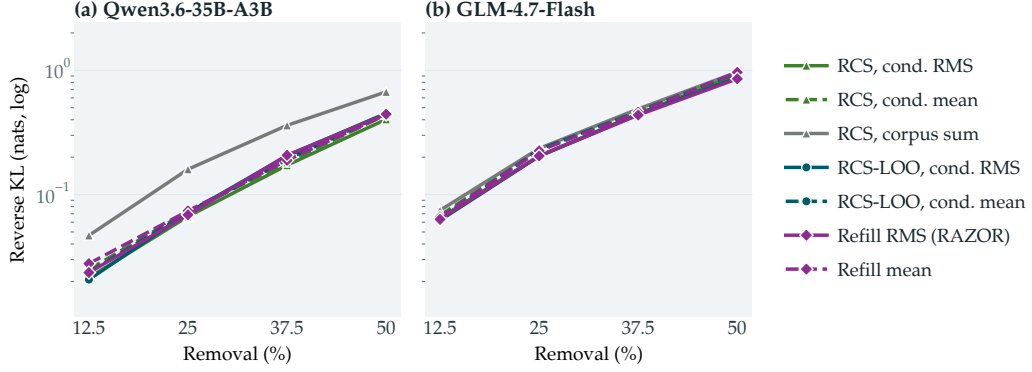


Figure 6: **Reverse KL under different calibration reductions (lower is better).** The RCS quantity uses conditional RMS, conditional mean, and corpus sum. The RCS-LOO and RCS-Refill quantities each use the two conditional reductions. Selection frequency is excluded because it discards the token quantity rather than reducing it.

These aggregate statistics do not identify ρ_i or separate the contributions of direction and residual magnitude. They also cannot recover the nonlinear factors, because the score ratios summarize experts whereas the routing statistics summarize routed events.

Damage-profile similarity and selection. Treating δ_i^{loo} and δ_i as vectors over each expert’s routed tokens, their cosine has median 0.9942 on Qwen3.6-35B-A3B and 0.9952 on GLM-4.7-Flash, with fifth percentiles 0.9630 and 0.9715. Most profiles are therefore close in direction, although the minima of 0.7062 and 0.5440 show substantial changes for some experts. Directional similarity alone does not fix the ranking, since nearly proportional profiles with different scale factors still reorder experts. The direction of the median score change does not determine the KL change. Despite the opposite median score shifts on the two models, refill lowers KL at both Qwen3.6-35B-A3B budgets and at only one GLM-4.7-Flash budget, so these local statistics do not predict its contribution to checkpoint quality.

C.5 REFERENCE, ROUTING FACTOR, AND AGGREGATION

Table 9 reports fourteen scoring configurations for two models at 25% and 50% removal. Its core grid crosses two output references with two routing factors. Rows within each block compare token quantities, while blocks compare reductions of the same quantity. The conditional mean of $w_i \|f_i\|_2$ is REAP, and the conditional RMS of $\frac{w_i}{1-w_i} \|f_i - c\|_2$ is RCS-LOO. Adding refill damage gives five token quantities across the corpus-sum $\sum_t g_i(t)^2$, conditional-mean, and conditional-RMS blocks, giving fourteen rows per model.

One row is not exactly matched to the rest. The conditional-mean refill row comes from the main sweep of Figure 6 rather than this grid’s evaluation collection, so comparisons involving it are cross-collection contrasts, unlike the matched comparisons among the other thirteen rows.

Evaluation and aggregation. Within each model, thirteen of the fourteen rows share four evaluation shards with matched domain sequences and token counts. Let $L_{s,d}$ denote mean reverse KL in shard s and domain d , with $n_{s,d}$ scored tokens. The estimator pools tokens within domains and then weights the eight domains equally,

$$L = \frac{1}{8} \sum_{d=1}^8 \frac{\sum_{s=0}^3 n_{s,d} L_{s,d}}{\sum_{s=0}^3 n_{s,d}}. \quad (22)$$

The entries are shard-based point estimates, not estimates of variability across independent runs.

Relation to the matched component study. The factorial and matched component studies use overlapping evaluation data but differ in token coverage within one of their four subsets. We therefore

Table 8: Calibration statistics of routing weights and refill-induced score changes.

Quantity	Qwen3.6-35B-A3B	GLM-4.7-Flash
<i>Routing geometry (token-weighted)</i>		
Selected weight w_i ($= 1/k$)	0.1250	0.2500
Promoted pseudo-weight w_r	0.0740	0.1325
$\cos(r_i, r_r)$	0.1080	0.1034
<i>Score ratio $s_i^{\text{rf}} / s_i^{\text{loo}}$ (per expert)</i>		
Median	1.0281	0.9744
Experts with ratio > 1	85.2%	14.3%
<i>Token-axis $\cos(\delta_i^{\text{loo}}, \delta_i)$ (per expert)</i>		
Median	0.9942	0.9952
p_5	0.9630	0.9715
Minimum	0.7062	0.5440
<i>Collection</i>		
Top- k / experts per layer	8 / 256	4 / 64
Layers / scored experts	40 / 10,240	46 / 2,944
Routed token positions	785,703	785,632

Table 9: **Fourteen scoring configurations varying reference, routing factor, and aggregation.** Reverse KL is in nats at 25% and 50% removal, with lower values better. Thirteen rows share four evaluation shards. The conditional-mean refill row comes from the main sweep in Figure 6, which differs in one shard. Comparisons involving this row are not fully matched. Sum denotes $\sum_t g_i(t)^2$. No corpus-sum result is available for $\frac{w_i}{1-w_i} \|f_i\|_2$. Refill damage with RMS is RAZOR.

Token quantity	Aggregate	Qwen3.6-35B-A3B		GLM-4.7-Flash	
		25%	50%	25%	50%
$w_i \ f_i\ _2$	Sum	0.14712	0.67725	0.22309	0.89425
$w_i \ f_i - c\ _2$	Sum	0.16005	0.67173	0.23513	0.96880
$\frac{w_i}{1-w_i} \ f_i - c\ _2$	Sum	0.17659	0.69294	0.23339	0.91110
$\frac{\ w_i r_i - w_r r_r\ _2}{1-w_i+w_r}$	Sum	0.18798	0.70264	0.23176	0.88086
$w_i \ f_i\ _2$	Mean	0.07469	0.45691	0.22852	1.01692
$\frac{w_i}{1-w_i} \ f_i\ _2$	Mean	0.07882	0.48975	0.22922	0.97162
$w_i \ f_i - c\ _2$	Mean	0.07357	0.44075	0.22804	0.94483
$\frac{w_i}{1-w_i} \ f_i - c\ _2$	Mean	0.07151	0.45341	0.23017	0.95748
$\frac{\ w_i r_i - w_r r_r\ _2}{1-w_i+w_r}$	Mean	0.07386	0.44143	0.22481	0.96456
$w_i \ f_i\ _2$	RMS	0.06869	0.43948	0.20813	0.86866
$\frac{w_i}{1-w_i} \ f_i\ _2$	RMS	0.09535	0.64640	0.20865	0.90296
$w_i \ f_i - c\ _2$	RMS	0.06674	0.40355	0.20483	0.92318
$\frac{w_i}{1-w_i} \ f_i - c\ _2$	RMS	0.07144	0.45134	0.20353	0.87107
$\frac{\ w_i r_i - w_r r_r\ _2}{1-w_i+w_r}$	RMS	0.06874	0.44409	0.20461	0.85460

report each study separately and do not treat their agreement as an independent replication. The mean-refill row is explicitly identified as a cross-study comparison.

No single scoring component is uniformly beneficial. The effect of each component depends on the other scoring choices. Even with mean aggregation fixed, the effect of the routing factor depends on the output reference. On GLM-4.7-Flash at 50% removal, replacing w_i with $w_i/(1-w_i)$ lowers reverse KL from 1.01692 to 0.97162 for output magnitudes, but raises it from 0.94483 to 0.95748 for consensus residuals. For fixed-support deletion damage, RMS lowers reverse KL relative

to the conditional mean in all four displayed model–budget settings, although the Qwen3.6-35B-A3B differences are small.

The reduction ordering depends on the token quantity. The grid reproduces the RCS ordering $\text{RMS} < \text{mean} < \text{sum}$ from Appendix C.3 at both displayed budgets on both models, but this ordering does not hold for every quantity. Conditional RMS has the lowest KL in 15 of the 16 cells with all three reductions, including four cross-collection comparisons involving mean refill. The exception is refill at 50% removal on Qwen3.6-35B-A3B. On GLM-4.7-Flash, corpus sum yields lower KL than conditional mean for output magnitudes at both budgets (0.22309 versus 0.22852, and 0.89425 versus 1.01692) and for both residual quantities at 50% removal (0.91110 versus 0.95748, and 0.88086 versus 0.96456), reversing the middle two terms in four of the sixteen cells. On Qwen3.6-35B-A3B, mean yields lower KL than sum in all eight available comparisons, with sum yielding $1.48\text{--}2.55\times$ the KL of mean. RMS beats sum in all sixteen matched comparisons, but the mean–sum ordering depends on the quantity and backbone.

The reference and the reduction interact. For the four non-refill quantities in the core grid, RMS lowers reverse KL relative to mean in 14 of the 16 quantity–model–budget cells. Both exceptions are the routing factor without the consensus reference, $\frac{w_i}{1-w_i} \|f_i\|_2$, on Qwen3.6-35B-A3B, where RMS raises KL from 0.07882 to 0.09535 at 25% removal and from 0.48975 to 0.64640 at 50%. Read in the other direction, the benefit of consensus subtraction depends on which reduction is in use. With the routing factor held at w_i , subtracting the consensus lowers KL in all four settings under the mean but in three under RMS, raising it from 0.86866 to 0.92318 on GLM-4.7-Flash at 50% removal. With the leave-one-out factor, it lowers KL in all four settings under RMS but in three under the mean, raising it from 0.22922 to 0.23017 on GLM-4.7-Flash at 25% removal. The two components are therefore not separable additive improvements.

An aggregation-matched baseline narrows the residual advantage. Comparing our criteria against REAP confounds the reference with the reduction, so the grid also aggregates REAP’s token quantity by conditional RMS. Under that matched baseline, REAP-RMS gives lower KL than RCS-LOO in both Qwen3.6-35B-A3B settings and slightly lower KL on GLM-4.7-Flash at 50%, so the benefit of RMS over mean aggregation for fixed-support deletion damage does not imply that the complete RCS-LOO criterion outperforms REAP-RMS. This comparison changes both the output reference and the survivor-renormalization factor. RCS-Refill fares better against the same baseline. It attains the lowest KL of all fourteen rows on GLM-4.7-Flash at 50% (0.85460) and beats REAP-RMS on GLM-4.7-Flash at both budgets, while trailing it on Qwen3.6-35B-A3B at both budgets (0.06874 versus 0.06869 and 0.44409 versus 0.43948), where RCS remains lowest. With both quantities aggregated by the conditional mean instead, refill yields lower KL than REAP in all four settings (0.07386 versus 0.07469 and 0.44143 versus 0.45691 on Qwen3.6-35B-A3B, 0.22481 versus 0.22852 and 0.96456 versus 1.01692 on GLM-4.7-Flash), although mean RCS-LOO still edges out mean refill on Qwen3.6-35B-A3B at 25% (0.07151). These mean-refill contrasts use the separate main-sweep shards rather than the identical batches used elsewhere in this table.

Taken together, the grid shows that local exactness of a token quantity does not determine its empirical ordering after joint pruning, which is the same backbone dependence visible in the main-sweep comparison of Table 6. This is why the choice of RAZOR rests on the benchmark results rather than on these checkpoint-fidelity contrasts, which do not identify the source of the downstream gains.

C.6 MATCHED COMPONENT CONTRASTS AT TWO REMOVAL BUDGETS

The matched component study uses four evaluation subsets at 25% and 50% removal, pools tokens within each axis, weights the eight axes equally, and varies one scoring choice at a time with the others held fixed. This separates the effects of the output reference, survivor renormalization, refill, and aggregation under a common protocol.

Consensus subtraction lowers reverse KL in every matched setting. With routing weights and mean aggregation fixed, it does so by 1.5%/3.5% on Qwen3.6-35B-A3B and 0.2%/7.1% on GLM-4.7-Flash at 25%/50% removal. With residuals and the LOO factor fixed, RMS likewise lowers KL in all four settings, by 11.6%/9.0% on GLM-4.7-Flash but by only 0.05%/0.44% on Qwen3.6-35B-A3B, which is too small a margin to read as separation. Beyond these two components the picture splits

by backbone. Relative to RMS without LOO, RCS-LOO lowers GLM-4.7-Flash’s reverse KL but raises Qwen3.6-35B-A3B’s at both budgets (Table 6). For refill damage, RMS changes reverse KL by $-7.0\%/+0.6\%$ on Qwen3.6-35B-A3B against $-9.0\%/-11.4\%$ on GLM-4.7-Flash.

The two fidelity metrics also disagree, which matters because each is a defensible summary of the same predictions. In the RMS contrast at fixed residuals and LOO factor, RMS raises Qwen3.6-35B-A3B’s excess perplexity by $16.5\%/2.7\%$ while lowering its reverse KL, whereas it lowers GLM-4.7-Flash’s by $12.5\%/13.0\%$ in agreement with KL. The refill contrast shows the same split, changing excess perplexity by $-8.5\%/+6.9\%$ on Qwen3.6-35B-A3B and $-8.9\%/-13.2\%$ on GLM-4.7-Flash. Appendix E.4 examines this disagreement directly on shared routing groups. Here it means that no ordering of the proxies holds under both metrics at once.

Whether the gains reach beyond calibration-covered task types is a separate question, which we address by averaging the four covered and four held-out axes separately with equal weight within each group, using excess perplexity $\exp(\Delta\text{NLL}) - 1$. REAP’s held-out reverse KL is $2.1\text{--}2.9\times$ its in-domain value across the two budgets and its held-out excess perplexity is $3.4\text{--}3.8\times$ its in-domain value. Domain-specific comparisons appear in Appendices E.1 and E.2.

Across all eight axes, GLM-4.7-Flash improves on both metrics, with RCS-LOO reducing KL by $10.9\%/14.3\%$ and excess perplexity by $8.9\%/17.1\%$ relative to REAP, and RAZOR reducing them by $10.5\%/16.0\%$ and $8.7\%/19.3\%$. Qwen3.6-35B-A3B again splits, as RCS-LOO lowers its all-axis KL by $4.3\%/1.2\%$ while changing excess perplexity by $-0.7\%/+12.6\%$, and RAZOR lowers KL by $7.9\%/2.8\%$ while changing excess perplexity by $-9.5\%/+6.6\%$. Refill is the one component that helps that model on both metrics, lowering its excess perplexity by $8.9\%/5.3\%$ relative to RCS-LOO.

C.7 SELECTION DIAGNOSTICS

The comparisons above measure score values and checkpoint fidelity, neither of which shows what changed in the retained set. Three probes address that question from different angles. Conditioning on keep-set disagreements asks whether the tokens where two criteria retain different experts are where their outputs diverge. Joint residual statistics ask whether the residuals inside a selected deletion set cancel or reinforce. Single-expert ablations ask whether local score order survives deployment, once the router reselects from the remaining pool. These probes characterize selection and local interactions. They do not measure redundancy among the final retained experts or establish the mechanism behind the benchmark gains.

C.7.1 OUTPUT DIFFERENCES CONDITIONED ON SELECTION DISAGREEMENTS

We compare RCS-LOO and REAP at 50% removal on 16 batches spanning eight axes for each of Qwen3.6-35B-A3B and GLM-4.7-Flash. This collection is separate from the four-shard main sweep and is subject to the checkpoint-matching limits in Appendix B.5. It contains no RAZOR predictions, so this probe is not repeated for refill.

Each conditioning variable uses routed events from a specified layerwise keep-set disagreement. For routing concentration, eligible experts are retained by RCS-LOO but not by RCS. We select the event with the largest $1/(1 - w_i)$ across layers for each token and use its normalized routing entropy. For output consensus, eligible experts are retained by residual RMS without LOO but not by REAP. We select the event with the largest residual damage relative to the expert’s calibration mean and use its $\kappa = \|c\|_2^2 / \sum_j w_j \|f_j\|_2^2$ and relative residual norm. For relative impact, eligible experts are retained by RCS-LOO but not by conditional-mean fixed-support scoring. We take the largest $\delta_i^{\text{loo}} / \mathbb{E}_{\text{cal}}[\delta_i^{\text{loo}} \mid i \in S]$ across their routed events, or zero if none is observed. This ratio measures shift relative to an expert’s mean damage, not event rarity or preservation of rare capabilities.

All diagnostics use model-wide quantiles on a common finite-token subset. Within each bin, we average RCS-LOO-minus-REAP reverse KL within batches and then equally across batches, with 5,000 paired-batch bootstrap resamples. The most concentrated routing, highest-consensus, and highest-relative-impact bins give Qwen3.6-35B-A3B and GLM-4.7-Flash differences of $-0.14/ -0.33$, $-0.18/ -0.28$, and $-0.25/ -0.47$, respectively. Tokens where disagreeing experts carry the most concentrated routing, highest consensus, or highest relative impact tend to show the largest criterion gaps. These associations are conditional on selection disagreements and do not generalize to unconditioned tokens.

C.7.2 INTERACTIONS WITHIN THE SELECTED DELETION SETS

The joint identity in Eq. 4 motivates a separate test of whether cross-expert residuals cancel or reinforce within score-selected deletion sets. We select sets with RCS-LOO at 25% and 50% removal and accumulate their interaction Gram matrices on the diagnostic calibration collections of Appendix B.1. Define $u_i(x) = w_i(x)(f_i(x) - c(x))$ for routed experts and zero otherwise, and let $G_{ij} = \sum_x \langle u_i(x), u_j(x) \rangle$ within a layer. For deletion set $\mathcal{A}_\ell$, the interaction ratio is

$$\gamma_\ell = \frac{\sum_{i,j \in \mathcal{A}_\ell} G_{ij}}{\sum_{i \in \mathcal{A}_\ell} G_{ii}}, \quad (23)$$

provided the denominator is positive. The common layer output scale cancels in the ratio. Values below one indicate net cancellation, while values above one indicate reinforcement. We compute one ratio per layer and report equal-layer quantiles over 40 Qwen3.6-35B-A3B and 46 GLM-4.7-Flash layers.

At 25% and 50% removal, median ratios are 1.080/0.921 on Qwen3.6-35B-A3B and 0.992/0.811 on GLM-4.7-Flash. At 50%, 29 of 40 Qwen3.6-35B-A3B layers and all 46 GLM-4.7-Flash layers show net cancellation. Sets selected by RAZOR from the same accumulators give median ratios of 1.077/0.925 on Qwen3.6-35B-A3B and 0.997/0.831 on GLM-4.7-Flash, with cancellation in 29 of 40 and 42 of 46 layers at 50%. The degree of cancellation is therefore similar across criteria at both budgets. These ratios characterize only the squared numerator of the fixed-support perturbation. They omit the token-dependent factor $(1 - W_{\mathcal{A}}(x))^{-2}$ and router refill, and the identity $\sum_i u_i(x) = 0$ constrains their cross terms. They therefore do not measure joint deletion loss. The fixed-support counterfactual is undefined when no routing mass survives.

C.7.3 SINGLE-EXPERT REMOVAL WITH DEPLOYED ROUTING

This probe tests whether local saliency transfers to end-to-end single-expert damage under deployed routing. We uniformly sample 32 experts and intervene on each in six layers spanning shallow, middle, and deep positions, giving 192 ablations per model. Each intervention masks one expert’s router score to $-\infty$ and reselects and renormalizes the top- k from the remaining pool, potentially admitting previously inactive experts.

We compare the intervened model q with the unpruned model p on 16 shared batches, two per evaluation axis, using assistant-token reverse KL $D_{\text{KL}}(q||p)$. The resulting damage estimate pools 78,091 scored tokens on Qwen3.6-35B-A3B and 79,975 on GLM-4.7-Flash, weighting tokens rather than axes equally. For each scoring rule, we compute Spearman correlation over the 32 interventions within each layer and average the six correlations equally. The saliency statistics come from the diagnostic calibration collections of Appendix B.1.

Mean correlations for RCS-LOO, REAP, and EAN are 0.31, 0.34, and 0.31 on Qwen3.6-35B-A3B, and 0.19, 0.18, and 0.10 on GLM-4.7-Flash, respectively. Every Qwen3.6-35B-A3B correlation is positive, but several GLM-4.7-Flash values are near zero and EAN is negative in two layers. The refill score RAZOR gives 0.31 on Qwen3.6-35B-A3B and 0.18 on GLM-4.7-Flash. The point estimates show no consistent advantage across models. This probe measures how well local output shifts under single-expert removal transfer to sample-wide prediction KL after router reselection. Better ordering of all single deletions does not guarantee better budgeted selection, and the probe accounts for neither the fixed-support identity nor joint deletion sets.

D CALIBRATION STABILITY AND SENSITIVITY

We examine how expert selection changes with calibration budget and how performance varies across calibration corpora. The budget study compares expert sets selected from disjoint token subsets for REAP and all three residual criteria. The corpus study evaluates reference-token PPL for 24 RCS-LOO checkpoints. Its configurations also differ in token budget and conversation structure, so the PPL contrasts do not isolate corpus choice.

D.1 CALIBRATION BUDGET AND SELECTION STABILITY

The Qwen3.6-35B-A3B collection contains 32 shards and 2,151,112 tokens. At each budget, we draw pairs of disjoint subsets with equal shard counts, merge statistics within each subset, and select experts. Agreement is the sum of layerwise retained-set intersection sizes divided by the total number of retained experts across scored layers. We report averages over four subset-pair draws. Additive sufficient statistics allow scores to be recomputed for each subset without additional model evaluations. All four criteria use the same subsets. Subset agreement compares the two disjoint selections. Full-set agreement compares each selection with that from all 32 shards and averages the two values. Only the former measures agreement from disjoint calibration data, because the full-set reference includes both subsets.

Selection agreement generally increases with calibration budget for all four criteria (Figure 7). For RCS-LOO, disjoint-subset agreement rises from 85.72% to 93.39% at 50% removal as the budget grows from approximately 65,500 to 1.08 million tokens. At 25% removal, it rises from 92.82% to 96.58%. In the final budget step, disjoint-subset agreement increases by only 0.01 percentage points for RCS-LOO at 50% removal. At 25%, the corresponding changes across criteria range from -0.03 to $+0.07$ percentage points. These small changes across four draws do not establish whether agreement has plateaued.

Across eight calibration budgets, two removal ratios, and two agreement measures, REAP has the highest selection agreement and RCS the second highest in all 32 settings. RAZOR exceeds RCS-LOO in 31 settings. At the smallest budget and 50% removal, disjoint-subset agreement is 89.50% for REAP and 85.72% for RCS-LOO. At approximately 1.08 million tokens, their disjoint-subset gap narrows to 2.64 percentage points, alongside a narrowing full-set agreement gap. The residual criteria therefore produce less stable retained sets than REAP throughout this collection, and more calibration data reduces but does not close that difference.

Selection stability and task performance measure different properties. These means over four subset-pair draws do not establish whether the lower stability of residual criteria affects fidelity or downstream performance. That question requires evaluating the checkpoints selected at each calibration budget.

D.2 CORPUS CONFIGURATIONS AND EVALUATION COMPARABILITY

The composition study compares full RAZORCAL, a tool-filtered subset, self-generated text, WikiText, and two complementary halves of the filtered subset at 25% and 50% removal. Filtering retains 1,790 of 2,048 examples by excluding 249 tool-calling and nine coding conversations with a tool-role message before the final assistant turn. The complementary halves combine alternating shards, $\{0, 2\}$ and $\{1, 3\}$. The filtered corpus retains RAZORCAL’s mixed-source provenance and is not a human-authored control. Self-generated variants use the corresponding unpruned backbone to continue the prefix before the first assistant turn. Since later dialogue turns are not preserved, these variants change conversation structure as well as response content. For article-style language-modeling text, we use the raw WikiText-2 training split of curated Wikipedia articles (Merity et al., 2016).

Perplexity across configurations. We evaluate 24 RCS-LOO checkpoints, one per configuration, model and removal ratio. Each is scored on 128 sequences of 2,048 tokens, yielding 262,016 next-token targets. PPL exponentiates the mean reference-token NLL over all next-token positions, not only assistant turns. The unpruned references are 5.781 on Qwen3.6-35B-A3B and 3.563 on GLM-4.7-Flash. Comparisons across configurations are subject to the input-matching limits below.

Input matching. The Qwen3.6-35B-A3B full-RAZORCAL and self-generated runs share the original model’s tokenization and evaluation rows. Their conversation pool differs from the eight-axis fidelity pool. It contains the first 128 conversations in source order that fill the token window, each truncated to its first 2,048 tokens. On these matched inputs, self-generated calibration yields lower PPL than full RAZORCAL at both removal ratios (6.468 versus 7.544 at 25%, and 6.991 versus 9.028 at 50%). This contrast still changes calibration token budgets and conversation structure as well as response content.

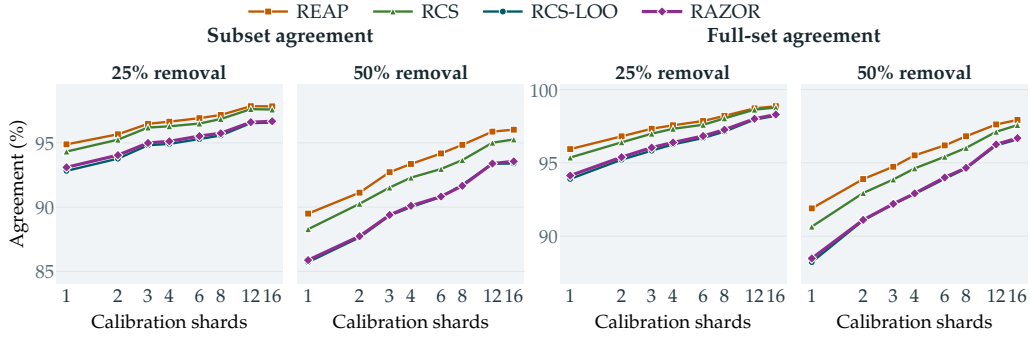


Figure 7: **Calibration budget and selection agreement on Qwen3.6-35B-A3B.** Subset agreement (left pair) compares two disjoint calibration subsets, while full-set agreement (right pair) compares each against the full 32-shard selection. Tokens per shard count are given in the text.

Across the larger collection, self-generated calibration yields lower PPL than tool-filtered calibration on Qwen3.6-35B-A3B but higher PPL on GLM-4.7-Flash at both budgets. WikiText calibration produces a 50%-removal Qwen3.6-35B-A3B checkpoint with PPL 4.831, below the unpruned reference. The complementary half-splits differ by 0.295 on Qwen3.6-35B-A3B at 25% removal and by 0.170 on GLM-4.7-Flash at 50%. Outside the matched Qwen3.6-35B-A3B contrast above, identical evaluation inputs have not been established across all configurations. Together with differences in calibration token budget and conversation structure, this limits interpretation of the PPL values as effects of calibration source. The study does not compare calibration robustness against REAP.

E OUTPUT FIDELITY AND RESPONSE DIVERSITY

We examine fidelity across evaluation axes and routing conditions, and compare it with response diversity and form. The analyses show domain-level regressions alongside aggregate KL gains (Appendix E.1), differences between KL and reference-token likelihood, and generation changes that do not follow the task-performance ranking.

Resampling units differ across analyses, so their intervals are not interchangeable. Per-axis fidelity resamples paired evaluation shards, domain-wise and routing-stratified analyses resample paired batches, and response diversity resamples questions. All reported 95% intervals are pointwise and unadjusted for multiple comparisons.

E.1 PER-AXIS FIDELITY RESULTS

The main sweep pools four disjoint stratified shards, weighting tokens within axes and the eight axes equally. Table 6 reports its absolute and relative reverse KL at 25% and 50% removal. The RCS-LOO checkpoints have lower reverse KL than REAP on both models at both budgets, with larger relative reductions on GLM-4.7-Flash.

At 25% and 50% removal, the axis-wise comparison separates four calibration-covered axes (*math*, *code*, *instruction following*, *tool use*) from four held-out axes (*chat*, *creative*, *safety*, *SQL*). RCS-LOO has lower reverse-KL point estimates in 11 of 16 axis-ratio cells on Qwen3.6-35B-A3B and 13 of 16 on GLM-4.7-Flash. To distinguish favorable and adverse directions, we enumerate all $4^4 = 256$ ordered resamples of the four paired shards. In each resample we recompute the token-weighted KL of both methods within the axis and take $100(L_{\text{method}}/L_{\text{REAP}} - 1)$. The 2.5th and 97.5th percentiles give pointwise intervals, of which 23 lie below zero, four lie above zero, and five include zero. Per setting, the below/above/includes-zero counts are 5/1/2 for Qwen3.6-35B-A3B at both budgets, 6/1/1 for GLM-4.7-Flash at 25%, and 7/1/0 for GLM-4.7-Flash at 50%. The four adverse cells are creative writing on Qwen3.6-35B-A3B at both budgets, SQL on GLM-4.7-Flash at 25%, and mathematics on GLM-4.7-Flash at 50%. Thus the aggregate improvement coexists with identifiable axis-level regressions. The same procedure applied to RAZOR gives lower point estimates in 11 of 16 cells on Qwen3.6-35B-A3B and 14 of 16 on GLM-4.7-Flash, with 24 intervals below zero, three above

zero, and five including zero (6/0/2 and 5/1/2 on Qwen3.6-35B-A3B at 25%/50%, 6/1/1 and 7/1/0 on GLM-4.7-Flash). Refill removes the adverse Qwen3.6-35B-A3B creative-writing interval at 25%, but the other three adverse cells remain. All comparisons share the same evaluation shards and ordered resamples within each axis. Their intervals are paired contrasts, not estimates of variation across checkpoints or calibration draws. This view therefore decomposes the same predictions rather than replicating the experiment.

E.2 DOMAIN-WISE OUTPUT FIDELITY

Figure 2 resolves fidelity by domain for four backbones at both removal ratios, using predictions collected by applying criterion-specific expert masks to each original model. These mask-based diagnostics are separate from the downstream benchmark runs. The figure compares REAP, RCS, RCS-LOO and RCS-Refill using reverse KL $D_{KL}(q||p)$ in nats over four calibration-covered and four held-out domains. Each domain estimate averages token means equally over six batches. ID/OOD summaries then give equal weight to domain means, and intervals use 10,000 paired-batch bootstrap draws, shared across criteria so every comparison stays paired on the same resampled batches.

The residual criteria improve on REAP in most domains, although the held-out results on Qwen3.6-35B-A3B are mixed. For RCS-LOO versus REAP, lower-KL domain counts are 6/4/7/6 at 25% removal and 7/5/7/7 at 50%, in GLM-4.7-Flash, Qwen3.6-35B-A3B, DeepSeek-V4-Flash-0731, and Hy3 order, each out of eight domains. Qwen3.6-35B-A3B’s held-out equal-domain mean is higher than REAP’s by 7.3% at 25% and 2.3% at 50%, with relative-change intervals crossing zero at both budgets, whereas the other three models have lower held-out means at 50%. Refill improves the point estimates in this group. For RAZOR the corresponding counts are 5/6/6/6 at 25% and 7/5/8/7 at 50%, and its Qwen3.6-35B-A3B held-out mean changes by -3.9% at 25% and $+0.2\%$ at 50% relative to REAP, again with intervals crossing zero and the other three models again lower at 50%. Held-out fidelity on that backbone is therefore where our advantage is weakest, and it is the same backbone whose excess perplexity moves against reverse KL in Appendix C.6.

Both removal ratios share reference predictions and token positions, with at most 1,024 valid positions per batch and no further subsampling. For RCS-LOO the 50% row uses the same predictions as Figure 8, but aggregates them by domain rather than routing decile. Neither view evaluates full generated trajectories.

Each spoke uses an independent KL scale spanning all four point estimates, padded by the median bootstrap half-width across criteria. Curves are comparable along a spoke, but distances are not comparable across spokes, panels, or removal ratios. Eight of 256 whiskers extend beyond the frame, so their upper ends are clipped rather than omitted. Intermediate labels give mid-axis KL values.

E.3 ROUTING-STRATIFIED OUTPUT FIDELITY

The LOO factor increases with an expert’s routing weight, motivating an analysis of whether residual criteria improve fidelity most under concentrated routing. Figure 8 groups assistant tokens by routing concentration within each learned-router MoE layer. We rank tokens by negative normalized routing entropy $-H_r$ and form ten quantile groups with shared ID/OOD boundaries. A common finite-value mask pairs methods and metrics, including the accompanying entropy and PPL measurements. We exclude layer–batch–decile cells with fewer than 20 tokens, then average within-cell token means equally over valid layers and batches. This grouped estimator differs from the main sweep’s equal-axis estimator. The horizontal axis runs from diffuse to concentrated routing.

Four calibration-covered and four held-out domains are evaluated separately. Within each ID/OOD group, 2,000 paired-batch bootstrap resamples preserve all ten deciles. The 25% and 50% curves share original-model routing profiles, token masks, quantile boundaries, and valid layer–batch–decile cells, but compute intervals separately.

Learned-router saliency scoring does not apply to DeepSeek-V4-Flash-0731’s three fixed-hash-router layers, which are excluded from the curves. Across criteria, these layers use identical frozen keep sets based on occurrence counts in the token-to-expert table, not calibration-token frequencies. Remapping preserves k distinct routes per token.

At 50% removal, held-out reverse KL is $2.1\text{--}3.0\times$ its corresponding in-distribution value. From the least to the most concentrated routing decile, RCS-LOO’s relative advantage widens by 15.6/17.2 percentage points on Qwen3.6-35B-A3B ID/OOD and by 4.3 points on GLM-4.7-Flash ID. Nominal paired-bootstrap intervals exclude zero for these three endpoint contrasts but cross zero for the other five, giving mixed evidence for a concentration-dependent advantage.

Aggregate point estimates favor RCS-LOO by 14.4% on GLM-4.7-Flash OOD, 15.4/10.3% on Hy3 ID/OOD, and 7.5/8.9% on DeepSeek-V4-Flash-0731 ID/OOD. Qwen3.6-35B-A3B’s OOD KL reduction is near zero (-1.5% , CI $[-5.3, +2.7]$). The same estimator applied to RAZOR, whose 50% predictions share the reference predictions and token mask, favors it by 15.6% on GLM-4.7-Flash OOD, 17.2/12.9% on Hy3 ID/OOD, and 10.0/13.1% on DeepSeek-V4-Flash-0731 ID/OOD, with Qwen3.6-35B-A3B OOD again nearly tied ($+0.5\%$, CI $[-3.7, +4.9]$). Relative to REAP, RCS-LOO’s mean absolute entropy drift is lower by 15%, 22%, 11%, and 21% on GLM-4.7-Flash, Qwen3.6-35B-A3B, DeepSeek-V4-Flash-0731, and Hy3, respectively. Unlike the KL reductions above, these entropy figures average over all eight axes rather than an ID or OOD group, so they do not describe held-out behavior.

Adding the leave-one-out factor does not uniformly improve routing-stratified fidelity under conditional RMS. Averaged over deciles, RCS attains the largest KL reduction against REAP in three of the eight model-group panels. These are both Qwen3.6-35B-A3B panels ($+15.9\%$ ID and $+11.9\%$ OOD, against $+9.4\%$ and -0.4% for RAZOR) and DeepSeek-V4-Flash-0731 ID ($+11.4\%$ versus $+9.7\%$). RAZOR leads the remaining five panels by 1.2–4.3 percentage points over the better of RCS and RCS-LOO. The largest RCS–RCS-LOO gap occurs on Qwen3.6-35B-A3B OOD, where RCS-LOO and RAZOR both fall below REAP on average, while RCS improves on it. The gap narrows at the tenth decile, where the reductions are $+16.5\%$, $+11.8\%$, and $+12.4\%$ for RCS, RCS-LOO, and RAZOR, respectively. These equal-decile means at 50% removal do not establish a downstream ranking or replace the matched equal-axis estimates in Table 6. The RCS–RCS-LOO contrast shows that the gate factor’s effect on fidelity depends on the backbone even with aggregation held fixed.

Entropy drift complements reverse KL by measuring changes in predictive dispersion. Predictive entropy itself is neither accuracy, calibration, nor response diversity, and neither higher nor lower entropy alone is preferable.

Reverse KL and perplexity deviation give different rankings. RAZOR lowers reverse KL relative to REAP in 73 of 80 deciles, with all seven exceptions in Qwen3.6-35B-A3B’s held-out group. On perplexity deviation, however, it is farther from the original than REAP in 7/10 ID deciles on Qwen3.6-35B-A3B and 9/10 on DeepSeek-V4-Flash-0731, while remaining closer in all twenty GLM-4.7-Flash ID and OOD deciles. Across all groups it preserves PPL more closely in 47 of 80 deciles. These are the same paired point comparisons summarized in Appendix E.4, not separate experiments.

The incremental effect of refill depends on the budget and domain group. At 50% removal, RAZOR exceeds RCS-LOO by 1.0–4.3 percentage points in REAP-relative KL reduction, averaging deciles equally within each of the eight model-domain groups. Their marginal bands overlap throughout on Qwen3.6-35B-A3B. At 25% removal, the ordering varies by model. In the calibration-covered groups, RAZOR leads by 8.2 points on Qwen3.6-35B-A3B but trails RCS-LOO by 3.7 on GLM-4.7-Flash and by 1.5 on DeepSeek-V4-Flash-0731. The refill margin also does not increase consistently with concentration, as its linear slope at 50% ranges from -0.30 to $+0.06$ percentage points per decile across the eight groups. Refill therefore has model- and budget-dependent effects, without a consistent trend across routing concentration.

E.4 REFERENCE-TOKEN PERPLEXITY ON THE SAME ROUTING GROUPS

To distinguish distributional fidelity from reference-token likelihood, we compute PPL at 50% removal for all four backbones on the same predictions, finite-token mask, and layer–batch–decile cells as Figure 8, whose lower two rows plot the result. Using the reference-token losses in Appendix B.4, let $\Delta\ell(t) = \ell_q(t) - \ell_p(t)$. Grouped PPL is

$$\text{PPL}_p = \exp(\mathcal{A}[\ell_p]), \quad \text{PPL}_q = \exp(\mathcal{A}[\ell_p + \Delta\ell]), \quad (24)$$

where $\mathcal{A}$ is the routing-stratified average over valid token cells, layers, and batches. Exponentiation follows NLL aggregation rather than preceding it. The result is neither mean per-token perplexity nor

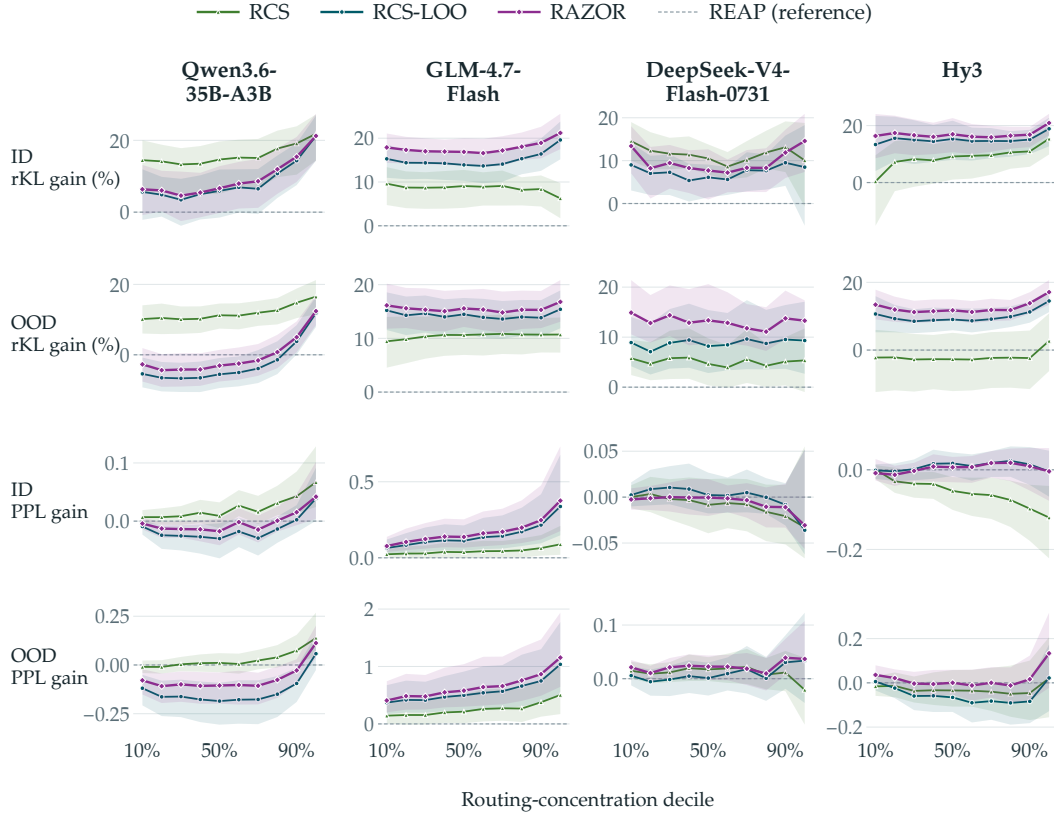


Figure 8: **Routing-stratified fidelity at 50% removal.** The three residual criteria use conditional RMS and are evaluated on the same batches, reference prefixes, and token positions. All panels compare against REAP at the same decile, with a dashed zero line and higher values indicating better fidelity. The top two rows show reverse-KL reduction in percent. The bottom two rows show REAP’s $|\text{PPL}_q - \text{PPL}_p|$ minus the criterion’s, in PPL points (Equation 24), so positive values mean closer to the original. Differences avoid unstable ratios where REAP’s deviation approaches zero (0.0016 on Hy3 ID). Bands are pointwise 95% paired batch-bootstrap intervals, and panels use independent vertical scales.

exponentiated predictive entropy, and need not equal whole-corpus token-weighted PPL. We measure preservation by the absolute deviation $|\text{PPL}_q - \text{PPL}_p|$, so “closer” means a smaller deviation than REAP on the same group, not simply a lower PPL.

The bootstrap pairs the original, REAP, and RCS-LOO predictions on the same retained positions. Within each ID/OOD group, shared whole-batch indices preserve all ten deciles. Recomputing and exponentiating each resampled NLL average gives marginal intervals and paired PPL differences.

The two fidelity views disagree widely. Across 80 point estimates from four models, two ID/OOD groups, and ten deciles, RCS-LOO has lower reverse KL than REAP in 72 but PPL closer to the original in only 47, and RAZOR gives 73 and 47 on the same cells. The disagreement is concentrated rather than diffuse. Both measures favor RCS-LOO throughout GLM-4.7-Flash’s groups, whereas on Qwen3.6-35B-A3B PPL is closer in only two ID and one OOD decile despite lower reverse KL in every ID decile. DeepSeek-V4-Flash-0731 illustrates why the measures can part ways, since its ID NLL changes are negative, so a lower PPL there means a larger deviation from the original rather than a smaller one. Improving the full predictive distribution and improving the likelihood of the particular reference continuation are therefore distinct objectives, and our scores optimize neither directly.

E.5 RESPONSE DIVERSITY

Fidelity on fixed reference prefixes does not determine diversity under free-running generation. We measure lexical and embedding diversity across repeated responses using two sampling protocols. A three-benchmark comparison covers all four criteria on Qwen3.6-35B-A3B, and a larger collection covers fewer criteria on four tasks and two backbones.

Criterion-family comparison. The left panel of Figure 3 compares REAP, RCS, RCS-LOO, and RAZOR on Qwen3.6-35B-A3B using Distinct- n for $n = 2, 3, 4$. For each benchmark, all four criteria at both budgets and the unpruned reference use 16 fixed response positions per question, drawn from the first 256 positions for AIME and 32 for each coding task, the largest pools common to every criterion. This comparison draws from different sampling pools than the four-task analysis below, so their percentages are not directly comparable.

Within each benchmark, values are $100(\bar{M}/\bar{M}_{\text{Original}} - 1)$ for question-averaged metrics. AIME 2026, HumanEval+, and LiveCodeBench receive equal weight, so AIME contributes one third of the weight on 30 of 369 questions. For pointwise 95% paired question-bootstrap intervals, we average the resampled relative changes across benchmarks before taking percentiles. These intervals condition on the 16 fixed responses rather than measure variation over new generations. Of twelve intervals per budget (four criteria and three orders), all twelve exclude zero at 50% and nine do so at 25%. They describe changes from Original, not the RCS-LOO-REAP difference. In per-benchmark point estimates, RCS-LOO exceeds REAP in all nine benchmark-order cells at 50% and six at 25%. Across the six benchmark-averaged budget-order cells, its median advantage over RCS is 4.02 percentage points in relative change from Original. Refill lowers Distinct- n relative to RCS-LOO in five of these six cells.

Four-task collection and sampling. The repeated-generation collection is separate from the nine-task benchmark. It contains 30 AIME 2026 (Mathematical Association of America, 2026), 219 Minerva (Lewkowycz et al., 2022), 164 HumanEval+ (Liu et al., 2023), and 175 LiveCodeBench (Jain et al., 2025) questions. The Minerva evaluation uses a 219-problem subset of the 272 OCWCourses undergraduate STEM problems assembled from MIT OpenCourseWare. The 175 LiveCodeBench task IDs are fixed across all variants compared here, which is what the paired comparisons require, and are drawn from a different release than the nine-task benchmark.

It covers Original plus EAN, Frequency, REAP, and RCS-LOO at 25% and 50% removal on Qwen3.6-35B-A3B, and Original, REAP, and RCS-LOO at both budgets on GLM-4.7-Flash. Unlike the three-benchmark comparison in Figure 3, this analysis includes Minerva and GLM-4.7-Flash.

Mathematics has 2,048 generations per question, HumanEval+ has 512, and LiveCodeBench has 256. Generation uses temperature 0.7, top- $p = 0.95$, top- $k = 20$, repetition penalty 1.0, an 8,192-token output cap, and thinking disabled, which differs from benchmark decoding in Appendix B.3. We sample 16 response positions per question without replacement, reusing the same positions across methods so that all comparisons are paired. This yields 131,712 responses. Of these, 53,136 Qwen3.6-35B-A3B responses cover the three tasks shared with Figure 3.

Metric definitions. Metrics are computed per question and then averaged over questions. We adapt the distinct- n approach of Li et al. (2016) to model tokens, using $n \in \{2, 4\}$ in the four-task analysis and $n \in \{2, 3, 4\}$ in Figure 3. Distinct- n divides the number of unique token n -grams pooled across 16 responses by their total occurrences, without crossing response boundaries. Jaccard distance averages $1 - |G_i \cap G_j|/|G_i \cup G_j|$ over all 120 response pairs, where G_i and G_j are response-level token 4-gram sets. Both lexical measures use each model family’s own tokenizer.

Embedding distance averages pairwise cosine distances using the fixed all-MiniLM-L6-v2 encoder.² Our full-response extension splits text into non-overlapping chunks of at most 254 WordPieces, mean-pools token representations, and normalizes each chunk vector. We then normalize their length-weighted mean. This retains content beyond the first chunk but not chunk order. The resulting distances do not measure execution behavior or mathematical-strategy equivalence.

²<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

Normalization and uncertainty. For each method, family, and task, this comparison reports $100(\bar{M}/\bar{M}_{\text{Original}} - 1)$ using question means $\bar{M}$, where Original denotes the unpruned MoE. Higher values indicate more diversity. We recompute this ratio over 2,000 paired question-bootstrap draws and report pointwise percentile 95% intervals, conditional on the 16 fixed responses and evaluated checkpoints. They exclude new-response and checkpoint-level variation.

Length control and interpretation. The four diversity metrics are correlated, so they are not independent evidence. We assess length sensitivity by repeating the analysis on exactly the first 128 tokens of each response. Within a family, we retain only questions eligible for every available method at both pruning ratios. In AIME 2026, Minerva, HumanEval+, and LiveCodeBench order, the resulting question counts are 30/219/134/143 for Qwen3.6-35B-A3B and 29/176/82/142 for GLM-4.7-Flash.

At 50% removal, the four-task mean Distinct-4 gap between RCS-LOO and REAP changes from +4.30% on full responses to -0.37% under prefix control on Qwen3.6-35B-A3B. The corresponding gaps on GLM-4.7-Flash are +10.35% and +16.50%. These means weight task-relative changes against REAP equally, unlike the Original-normalized diversity panel of Figure 3. The control changes both length and question eligibility, so Qwen3.6-35B-A3B’s reversal under prefix control reflects both factors and is not interpretable as a pure length effect. Lexical and embedding diversity measure surface variation in token sequences and do not measure correctness, token efficiency, or the number of distinct correct solutions.

E.6 RESPONSE DELIMITERS, FORMATTING, AND TERMINATION

We measure stray closing thinking delimiters, code-fence structure, and length-limited finishes. These observable response properties complement task scores without assessing semantic correctness. Their association with removal budget differs across backbones, and some change substantially at 75% removal.

Protocol and operational definitions. This analysis follows the repeated-generation protocol in Appendix E.5, using temperature 0.7, top- $p = 0.95$, top- $k = 20$, an 8,192-token cap, and thinking disabled. The tables and the three response-form panels of Figure 3 use 256 shared sample positions per question, giving 44,800 completions per variant on LiveCodeBench. These measurements characterize response form, not the benchmark accuracy of Table 1.

A completion contains a *stray* `</think>` when that literal string appears at least once. We count each such completion once, regardless of the number of occurrences. In the Qwen3.6-35B-A3B tokenizer it is a single vocabulary entry that is not flagged as a special token, so ordinary detokenization leaves it in the text. The chat template already closes the thinking block inside the prompt, so any further occurrence is emitted by the model and is not evidence about latent reasoning. *No fence* denotes the absence of a line-delimited code fence, while *prefix* denotes a nonempty prefix before such a fence. Both are shares of all completions, and neither is a semantic code or narration classifier. Unfenced responses can contain code, which the evaluator may extract through its bare-code fallback. *Length-limited* uses the decoder’s finish reason. The rightmost panel of Figure 3 reports a *loop share*, the fraction of length-limited completions in which fewer than one quarter of the token 4-grams are unique. The threshold is a heuristic definition of high repetition. The share measures how often length-limited completions also satisfy this criterion. It does not identify why generation reached the output cap or measure repetition among all completions.

Response lengths in this table count Unicode characters, whereas Figure 3 counts tokens under the backbone’s own tokenizer, matching the units of its Distinct- n panel. The ratio of median character length to median token length falls from 3.44 for the unpruned model to 3.04 for REAP at 50% removal. Consequently, the proportional increase in median character length is smaller than the increase in median token length. Intervals resample questions to retain dependence among completions sharing a prompt.

Sample-pool sensitivity. Rare-event estimates depend on the number of sampled completions. REAP’s stray-delimiter rate is 8.93 per 10,000 with 32 completions per question versus 13.39 with 256, although the criteria retain the same point-estimate ordering at 50% removal under either pool.

Table 10: **Response form on LiveCodeBench by criterion and budget, Qwen3.6-35B-A3B.** Rates are percentages except stray `</think>` per 10,000. Lengths are in kchar. Brackets give pointwise 95% question-bootstrap intervals. GLM-4.7-Flash is measured but not printed here because that collection covers only REAP and RCS-LOO, with no RAZOR row for comparison. Its RCS-LOO ladder appears in Table 11.

Criterion	Cut	Stray <code></think>/10k</code>	No fence	Prefix	Len-lim.	Med. kchar
<i>Qwen3.6-35B-A3B, 44,800 completions per checkpoint</i>						
Original	–	3.6 [1.6, 6.0]	7.1	61.4	16.0	8.3
EAN	25%	0.4 [0.0, 1.1]	8.7	52.3	19.5	9.7
Frequency	25%	2.0 [0.7, 3.6]	5.2	60.5	15.1	9.2
REAP	25%	1.6 [0.4, 3.1]	8.4	55.1	19.1	10.3
RCS-LOO	25%	0.2 [0.0, 0.7]	8.4	52.3	17.0	8.6
RAZOR	25%	0.4 [0.0, 1.1]	7.7	54.7	16.3	8.4
EAN	50%	4.2 [2.2, 6.9]	18.3	40.2	33.9	9.6
Frequency	50%	16.1 [10.9, 22.3]	2.9	30.4	13.0	6.7
REAP	50%	13.4 [7.6, 20.1]	11.5	66.3	24.2	12.0
RCS-LOO	50%	68.5 [22.8, 133.7]	10.2	16.6	25.1	7.3
RAZOR	50%	8.5 [4.2, 13.6]	8.2	19.6	25.8	7.8

Table 11: **The RCS-LOO budget ladder on LiveCodeBench.** Spearman correlations use the nine pruned checkpoints, not only the four shown and not the 0% reference. Units follow Table 10.

Quantity	0%	25%	50%	75%	ρ	p
<i>Qwen3.6-35B-A3B</i>						
Stray <code></think>/10k</code>	3.6	0.2	68.5	42.0	+0.75	0.020
No fence (%)	7.1	8.4	10.2	25.8	+0.82	0.007
Prefix (%)	61.4	52.3	16.6	43.6	−0.28	0.460
Length-limited (%)	16.0	17.0	25.1	42.0	+0.82	0.007
Median kchar	8.3	8.6	7.3	8.4	+0.43	0.244
<i>GLM-4.7-Flash</i>						
Stray <code></think>/10k</code>	0.4	0.0	2.9	70.5	+0.89	0.001
No fence (%)	0.2	0.5	1.7	17.0	+0.93	<0.001
Prefix (%)	66.5	68.9	58.3	14.3	−0.88	0.002
Length-limited (%)	17.5	19.6	18.6	78.7	+0.58	0.099
Median kchar	3.8	4.0	3.0	30.4	+0.17	0.668

Both criteria in the main-text panel use the full 44,800-completion pool, whose rates and intervals are reported below.

Budget-dependent changes. We evaluate the unpruned reference and nine RCS-LOO checkpoints with removal ratios up to 75% on each backbone. Table 11 shows the quarter budgets and reports each rank correlation over the nine pruned checkpoints, excluding the unpruned 0% reference, which is shown for comparison but is not a removal budget. Only RCS-LOO covers every measured budget in this collection, while Table 10 compares other criteria at shared budgets. On Qwen3.6-35B-A3B, the unfenced and length-limited shares each have Spearman $\rho = +0.82$ with removal budget ($p = 0.007$), and the rate of stray `</think>` has $\rho = +0.75$ ($p = 0.020$). GLM-4.7-Flash gives $\rho = +0.93$ ($p < 0.001$), $+0.58$ ($p = 0.099$), and $+0.89$ ($p = 0.001$), respectively. These descriptive associations are not stepwise monotonic. Among the undisplayed budgets, Qwen3.6-35B-A3B’s length-limited share falls at 40% and GLM-4.7-Flash’s stray-delimiter rate peaks at 60%. The displayed values also show Qwen3.6-35B-A3B’s stray rate higher at 50% than at 75%.

The remaining two measures show no such trend. Median response length has $\rho = +0.43$ ($p = 0.24$) on Qwen3.6-35B-A3B and $+0.17$ ($p = 0.67$) on GLM-4.7-Flash, and the prefix share has $\rho = -0.28$ ($p = 0.46$) and -0.88 ($p = 0.002$), respectively. Nonsignificance here is weak evidence either way, since all tests are exploratory, use nine pruned checkpoints on shared questions, report two-sided p -values uncorrected for multiple comparisons, and do not estimate independent-run variability.

At 75% removal, Qwen3.6-35B-A3B produces 25.8% unfenced and 42.0% length-limited completions, while GLM-4.7-Flash produces 78.7% length-limited completions at a median length of 30,444 characters. These rates identify a substantial change in termination behavior under the evaluated decoding protocol.

Refill and delimiter emissions on Qwen3.6-35B-A3B. At 25% removal, every criterion’s tag-rate point estimate is at or below the unpruned Qwen3.6-35B-A3B rate of 3.6 per 10,000, with RCS-LOO lowest at 0.2 [0.0, 0.7] (Table 10). At 50%, RCS-LOO reaches 68.5 [22.8, 133.7], versus REAP’s 13.4 [7.6, 20.1]. The refill checkpoint reduces the observed rate to 8.5 [4.2, 13.6], below REAP’s point estimate but with overlapping marginal intervals.

The RCS-LOO–REAP gap also appears when each variant’s generated responses are binned by their own lengths using cutoffs from the unpruned length distribution. In the fourth interval, rates are 100.7 and 14.2 per 10,000, with RAZOR at 14.4. Matching question and sample position against Original gives 307 newly occurring tag events for RCS-LOO, 60 for REAP, and 38 for RAZOR. The event comparison pairs question and sample position, whereas the length-stratified comparison uses common cutoffs with variant-specific bin membership. Neither isolates the effect of changed response length or termination. On the same collection, GLM-4.7-Flash’s corresponding RCS-LOO rate is 2.9 [1.1, 4.9], as shown at 50% removal in Table 11. REAP, which is not included in that table, has a rate of 1.6 [0.4, 2.7]. The magnitude of the Qwen3.6-35B-A3B gap therefore does not carry over. Across variants, 98–100% of tag-containing completions also contain code output, but this does not establish answer correctness or preserved utility.

Length shifts on Qwen3.6-35B-A3B. At every evaluated 25% Qwen3.6-35B-A3B checkpoint, the median paired length change relative to Original, matched by question and sample position, is at most 17 characters in magnitude. At 50%, REAP lengthens responses by a median 542 characters (67.3% of completions longer), RCS-LOO by 126 and RAZOR by 114, while Frequency shortens them by 129. The share of completions with a prefix before the code fence falls to 16.6% for RCS-LOO and 19.6% for RAZOR but rises to 66.3% for REAP, against 61.4% unpruned.

Task and backbone dependence. HumanEval+, the shortest task on both backbones, is the only task nearly free of visible tags throughout, with one tag-containing completion out of 671,744. On the other tasks, the two backbones differ. Qwen3.6-35B-A3B emits tags on LiveCodeBench but almost nowhere else, with one AIME 2026 completion out of 84,480 and none of 616,704 on Minerva. GLM-4.7-Flash instead shows this behavior on mathematics. At 50% removal, 2.90% [1.25, 5.18] of its AIME 2026 completions carry a tag under REAP and 7.25% [4.11, 11.22] under RCS-LOO. The corresponding rates are 0.67% and 0.54% on Minerva, against at most 0.03% on LiveCodeBench.

Task-level median length alone does not order these rates. Qwen3.6-35B-A3B’s AIME 2026 responses are longer than its LiveCodeBench responses (10,862 versus 8,332 characters) yet contain almost no tags. This comparison does not isolate length from other task differences.

Matched examples. For one LiveCodeBench question at a fixed sample position, the 50% checkpoints produce 1,894 and 1,897 characters under Frequency and RAZOR (both opening with code), 4,228 under REAP, 4,875 under EAN, and 6,923 under RCS-LOO, versus 3,423 unpruned. Only RCS-LOO emits a closing thinking tag, preceded by 6,233 characters ending with “Let’s code accordingly.” and followed by program text. This describes the visible response sequence, not a latent reasoning process. The median pre-tag length among this checkpoint’s tag-containing completions is 12,507 characters.

A separate RCS-LOO 70% completion reaches the cap after 21,966 characters without a code fence, contributing to both the unfenced and length-limited counts. The two examples were selected by median preamble length within their respective categories. The tag-containing example terminates normally. They illustrate the measured categories rather than establish comparative answer quality.

F EXTENDED RELATED WORK

MoE compression methods differ along three axes, namely the removal unit, the importance signal, and the selection or adaptation procedure applied afterwards. Sorting the literature this way clarifies what our comparison does and does not cover, because RAZOR varies only the importance signal. It removes whole experts, ranks them independently under a fixed layerwise budget, and does not update learned expert or router weights. The four subsections below group prior work by how far it departs from that setting, moving from alternative whole-expert saliency signals, through methods that model expert relations or search over candidate sets, to methods that change the surviving functions or the router itself, and finally to compression below the level of a whole expert.

F.1 WHOLE-EXPERT SALIENCY AND CONDITIONAL AGGREGATION

Routing, activation, and parameter statistics offer inexpensive rankings. SEER-MoE uses activation counts or routing-probability soft counts with layerwise or global removal (Muzio et al., 2024). AIMER instead ranks experts without calibration data using a normalized ℓ_1/ℓ_2 concentration statistic over their weights (Liu et al., 2026a). Jaiswal et al. (2025) compare several expert-dropping criteria, including activation norms. These frequency, activation-norm, and weight-concentration criteria do not reconstruct the routed mixture after deletion.

REAP, our closest magnitude-based comparator, averages $w_i \|f_i\|_2$ over tokens selecting expert i (Lasby et al., 2026). Its analysis includes promoted substitution and survivor renormalization, but its saliency uses only the removed expert’s weighted output norm. In Liu et al. (2026b), the exact token-level perturbation also contains both survivor reweighting and the promoted replacement. The authors describe direct candidate-wise rerouting as costly and report high overlap between selections induced by exact damage and its magnitude proxy. Our refill identity evaluates the token-level perturbation algebraically under the stated assumptions, using the original routed outputs and one promoted expert output. This requires additional local computation but avoids separately reconstructing every candidate pruned mixture. The fixed-support identity covers deletion without refill. Our conditional RMS aggregation differs from the corpus-summed damage used in that analysis.

Token-level scoring and cross-token aggregation remain separate choices. RAZOR uses conditional RMS rather than REAP’s conditional mean, while the component comparisons hold the token quantity fixed when varying aggregation (Appendix C). Neither conditioning nor RMS follows from the local deletion identity, and no aggregation rule is uniformly preferred by the available results.

F.2 EXPERT RELATIONS AND CANDIDATE-SET SELECTION

Relational information is not unique to consensus-residual scoring. STUN clusters router-derived behavior before representative selection or selective reconstruction and unstructured pruning (Lee et al., 2025). HC-SMoE instead clusters average expert outputs and merges each group (Chen et al., 2025). SHAPE uses routing co-occurrence in a coalition-utility proxy (Zhang, 2026). ConMoE combines routing-conditioned contribution with nearest-expert parameter distance before prototype reassignment (Yao et al., 2026). MAESTRO derives importance from cross-layer routing transitions before layerwise selection (Goel et al., 2026), with its main results using attention-only LoRA recovery while experts and routers remain frozen.

Our scores instead evaluate output change under specified deletion counterfactuals at a fixed layer input. The reference is the routed mixture, not a cluster representative or a routing graph. This distinction concerns the local score, not a claim that independent expert ranking captures all interactions. Residuals can reinforce or cancel under joint removal (Appendix A.3), so exact single-deletion quantities do not yield an exact set-selection algorithm.

Other approaches evaluate candidate sets directly. Lu et al. (2024) select retained combinations by layer-output reconstruction, EEP searches pruning and merging configurations (Liu et al., 2024), and MoE-I² evaluates layerwise deletion sets before jointly choosing candidates across short layer blocks (Yang et al., 2024). We instead retain the highest-scoring experts under a fixed layerwise budget. Our local single-deletion identities neither solve combinatorial selection nor establish superiority to subset search. Chowdhury et al. (2024) give theoretical conditions under a stylized classification

model in which pruning experts from a fine-tuned sparse MoE preserves performance. Our local surrogate does not provide such a guarantee.

F.3 MERGING, ROUTER ADAPTATION, AND BUDGET ALLOCATION

Expert merging changes the surviving functions, rather than only selecting which original experts remain. MC-SMoE groups experts around high-usage representatives using router-logit similarity, aligns neurons, and averages expert parameters before low-rank and structured compression (Li et al., 2024). REAM protects high-saliency experts as centroids and merges similar non-centroid experts into them (Jha et al., 2026). Neither operation is the router refill modeled by RCS-Refill, which promotes an existing unselected expert without averaging its parameters with the removed expert.

Router KD adapts only router parameters after compression, distilling the uncompressed model’s next-token distribution while freezing the experts and other backbone parameters (Hyeon & Do, 2026). It is a recovery procedure, not an expert-ranking criterion. Budget allocation is another distinct choice. GRAPE compares residual redundancy across layers to allocate non-uniform budgets, merging similar expert pairs within the selected layer (Zhang et al., 2026). MC-SMoE also permits adaptive layerwise merging ratios (Li et al., 2024). These examples show that merging, adaptation, and budget allocation are overlapping design choices rather than mutually exclusive method families. Our experiments use uniform layerwise removal ratios, leave retained expert weights unchanged, and perform no recovery training. Combining residual-based scores with merging, router adaptation, or non-uniform allocation is outside the present comparison, so our results do not establish their compatibility or joint benefit.

F.4 FINER-GRAINED COMPRESSION AND COMPARISON SCOPE

MoE-Pruner sparsifies weights within experts using weight magnitudes and router-weighted input activation statistics (Xie et al., 2024). HEAPr treats feed-forward intermediate neurons as atomic expert units and uses output-space second-order information to estimate their importance (Li et al., 2026). Both operate at a finer granularity than removing entire routed experts. An atomic expert in this terminology is not one of the router’s original selectable experts. Fisher-MoE also studies a finer-grained variant that ranks FFN intermediate dimensions by Fisher importance and reports capability concentration in small sets of those dimensions (He et al., 2026).

These approaches share an interest in preserving model behavior, but their removal units and compression procedures differ from ours, so their reported sparsity levels are not matched whole-expert budgets.

How pruned experts are judged is itself contested, which bears on our decision to report generation behavior alongside task scores. Jha (2026) prune two open-weight MoE backbones for coding under five selection strategies and report that the winning strategy changes between model families, and that perplexity can rate a broken model above an intact one. These findings concern cross-model sensitivity and disagreement between perplexity and functional performance. Our experiments examine related distinctions among predictive fidelity, downstream task scores, and generation behavior (Appendix E.4). That study evaluates recovery tuning separately, while our experiments use no recovery training and cover nine reasoning-intensive tasks.

Our comparisons hold the removal unit, layerwise budget, and learned parameters fixed while applying each architecture’s routing rule to the retained pool. They vary the scoring criterion among RCS, RCS-LOO, and RCS-Refill against the magnitude-based and frequency-based baselines, with RAZOR denoting RCS-Refill under conditional RMS. Our evidence therefore concerns which local signal best identifies replaceable experts, and it leaves open how such a signal would combine with merging, router adaptation, non-uniform budget allocation, or sub-expert granularity. Their joint benefit with residual-based ranking remains untested.